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**PARK et al.**

(54) **OSCILLATOR WITH LEAKAGE CURRENT COMPENSATION FUNCTION**

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| *H03K 3/011* | (2006.01) |

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CPC ............ ***H03K 3/0231*** (2013.01); ***H03B 5/04*** (2013.01); ***H03B 5/24*** (2013.01); ***H03K 3/011*** (2013.01)

(57) **ABSTRACT**

An oscillator includes a relaxation oscillating circuit including a first resistor-capacitor (RC) circuit generating a first oscillation voltage at a first node and a second RC circuit generating a second oscillation voltage at a second node, and configured to output an oscillation voltage through an output line coupled to an output node, and a voltage averaging feedback circuit configured to receive a reference voltage and the oscillation voltage to output a control voltage through an output terminal. The relaxation oscillating circuit includes a leakage current compensation circuit coupled to the output line and configured to provide a leakage compensation current to the output node.

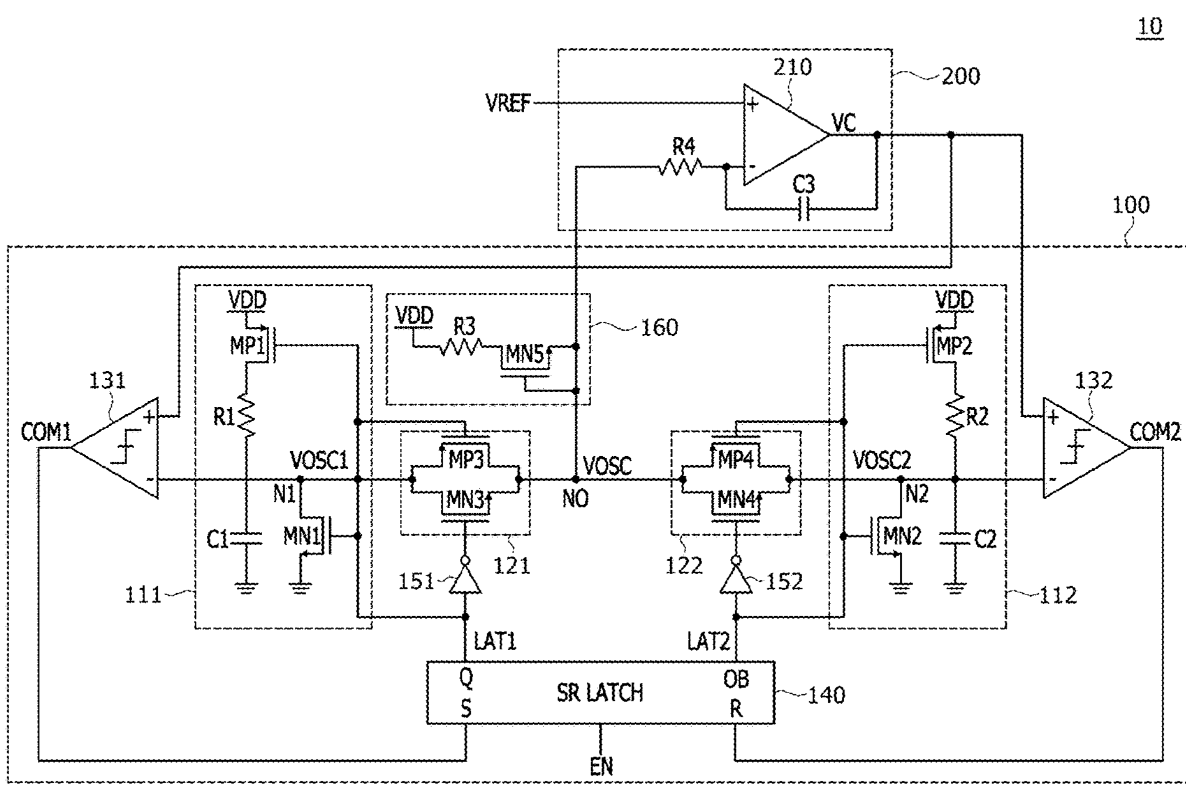
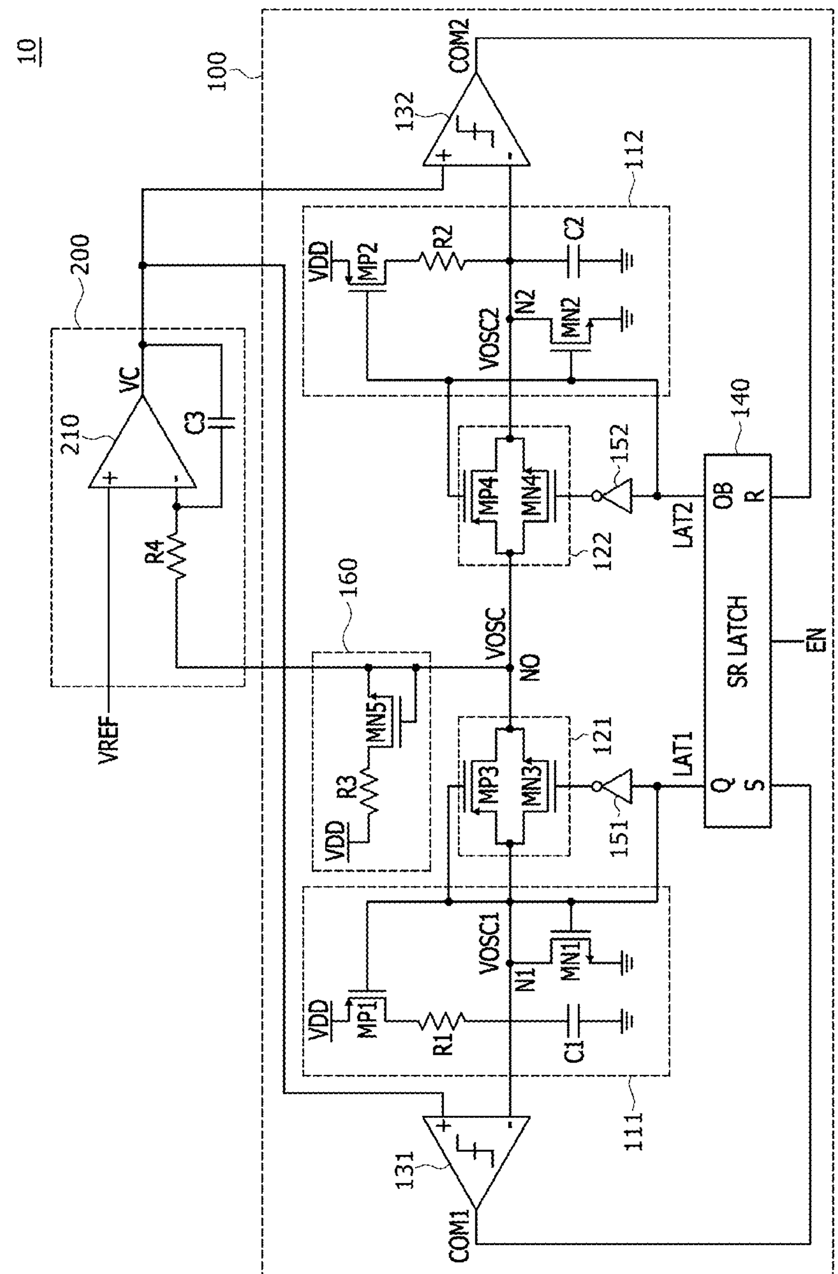


FIG.1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100
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VC
C3
R4
VREF
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VDD
R3
MN5
VOSC
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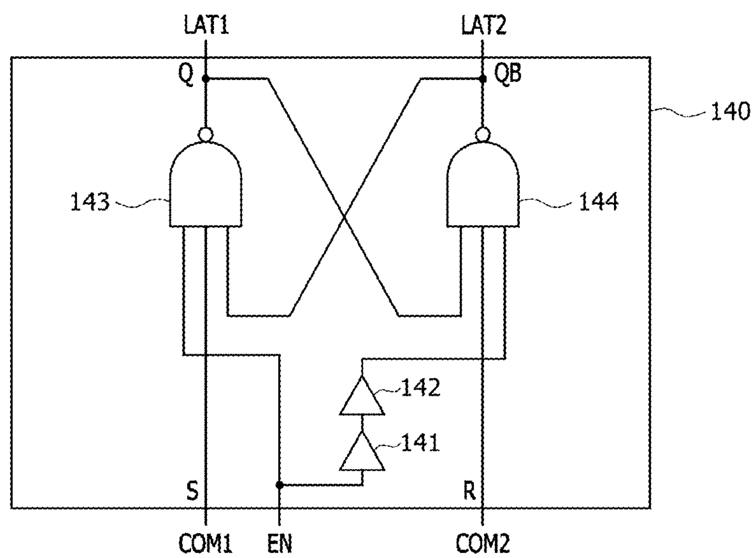
COM2
132
112
VDD
MP2
R2
C2
N2
VOSC2
MN2
MP4
MN4
152
122
LAT2
140
OB
R
SR LATCH
EN
Q
S
LAT1
121
MP3
MN3
151
111
VDD
MP1
R1
C1
N1
VOSC1
MN1
131
COM1

FIG.2



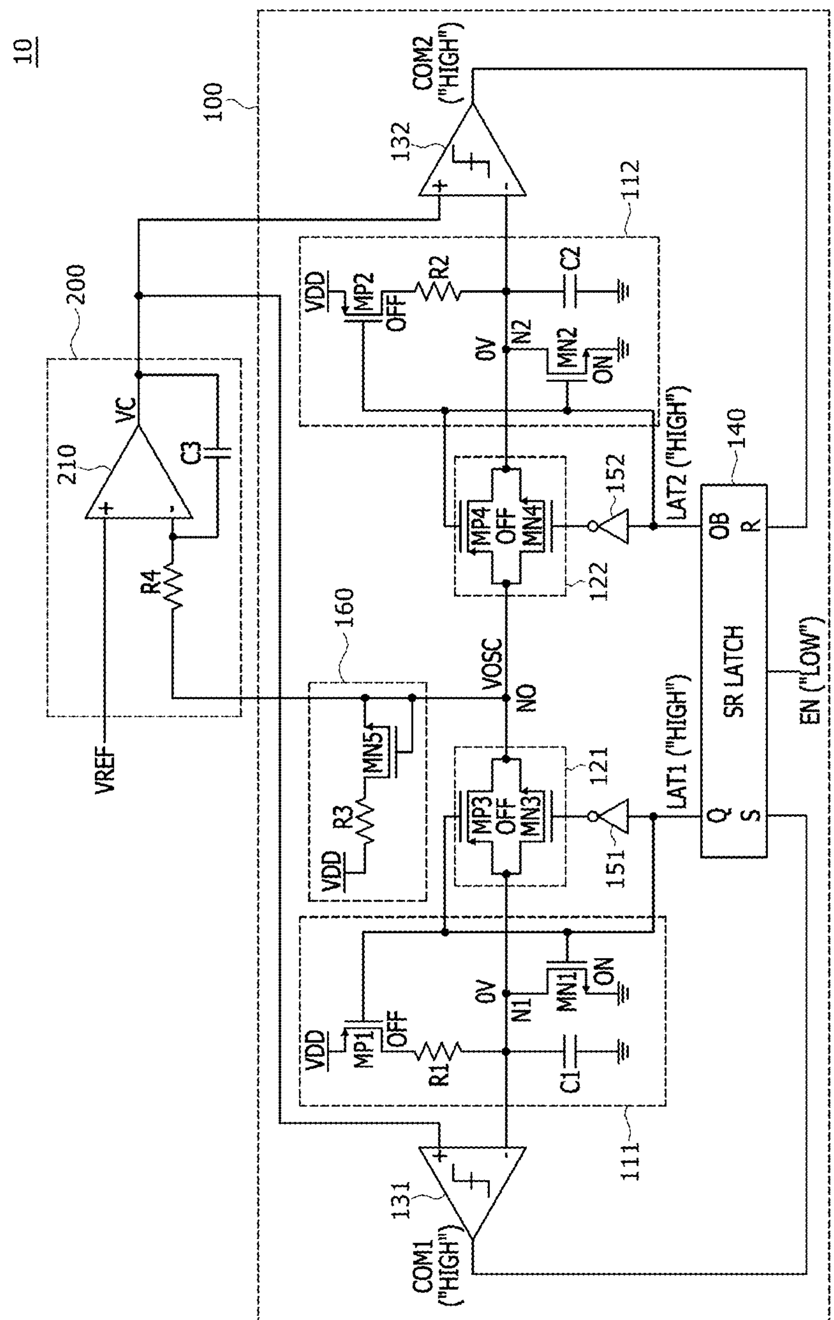
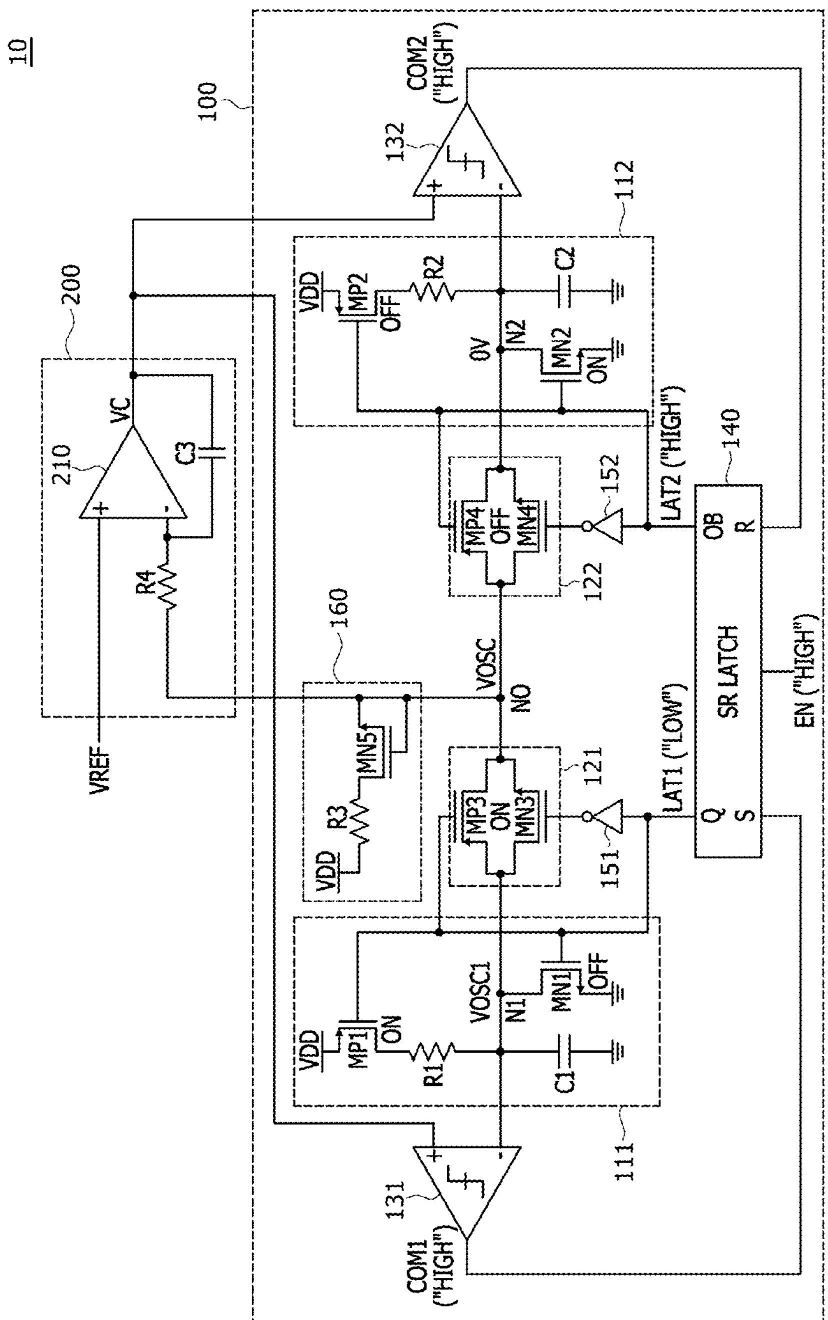

10
100
200
210
VC
C3
R4
VREF
160
MN5
R3
VDD
VOSC
NO
111
112
121
122
131
132
140
151
152
COM1 ("HIGH")
COM2 ("HIGH")
VDD
MP1
OFF
R1
C1
0V
N1
MN1
ON
VDD
MP2
OFF
R2
C2
0V
N2
MN2
ON
MP3
OFF
MN3
MP4
OFF
MN4
LAT1 ("HIGH")
LAT2 ("HIGH")
Q
S
QB
R
SR LATCH
EN ("LOW")

FIG.3

FIG.4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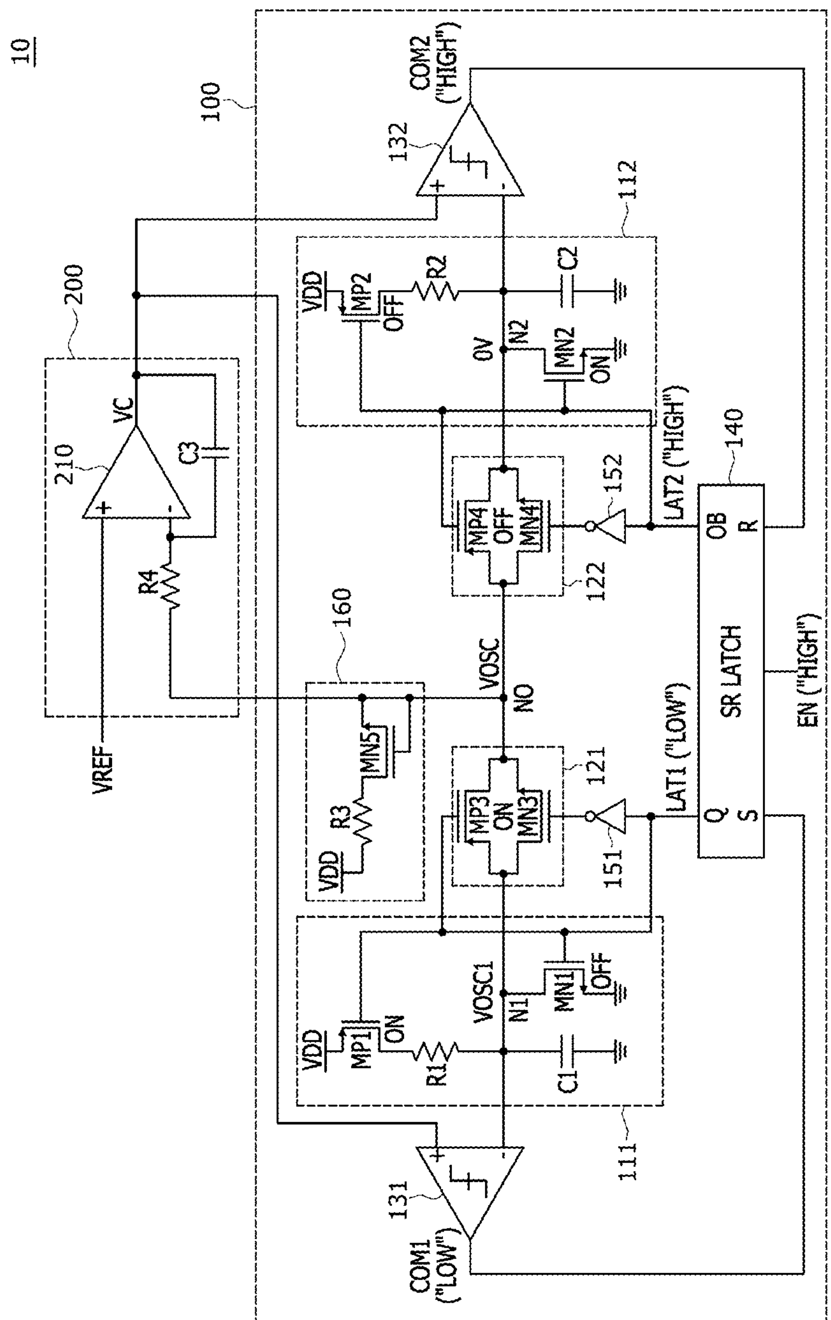
100
200
210
VC
C3
R4
VREF
160
VDD
R3
MN5
VOSC
NO
111
VOSC1
N1
MP1
ON
MN1
OFF
R1
C1
112
MP2
OFF
R2
0V
N2
MN2
ON
C2
121
MP3
ON
MN3
122
MP4
OFF
MN4
151
152
LAT1 ("LOW")
LAT2 ("HIGH")
140
SR LATCH
Q
S
QB
R
EN ("HIGH")
131
COM1 ("HIGH")
132
COM2 ("HIGH")

FIG.5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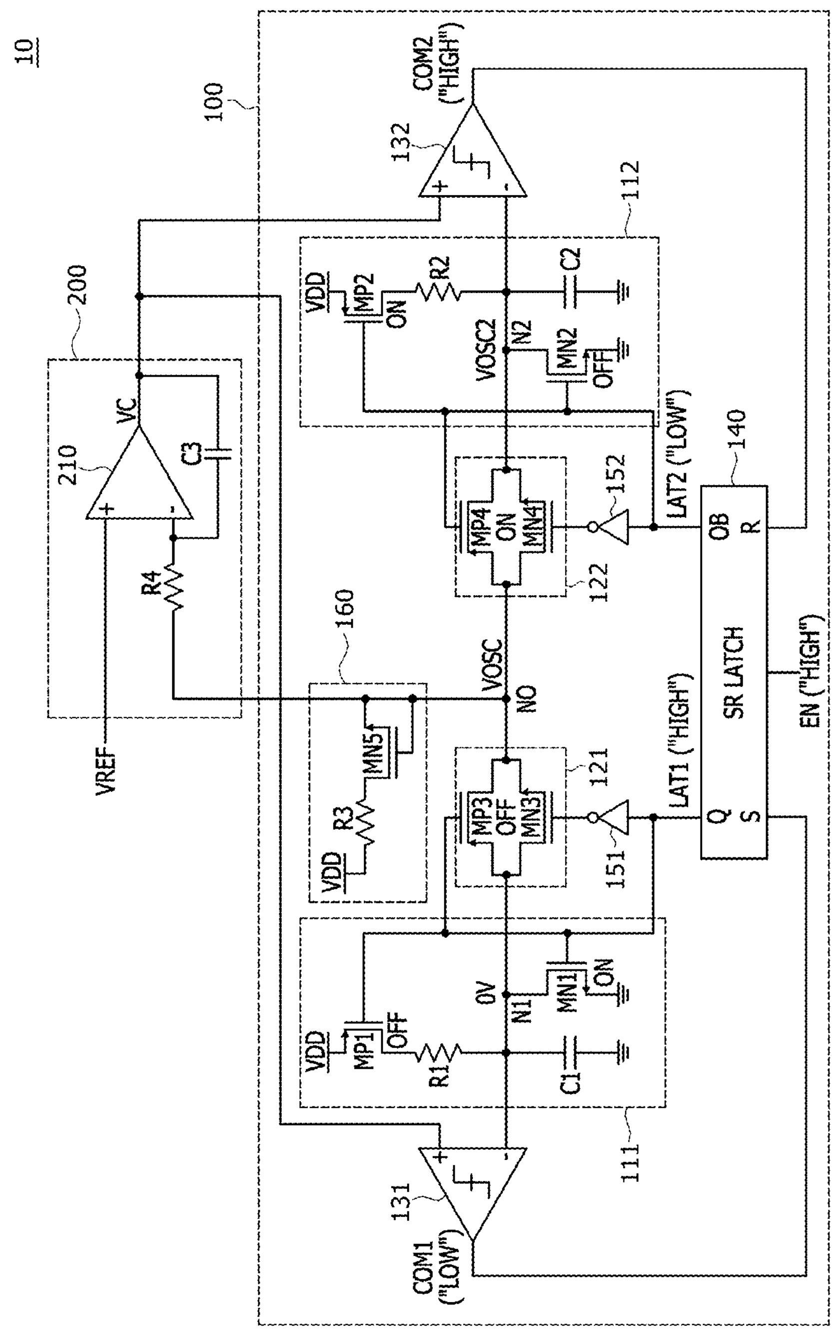
100
200
210
VC
C3
R4
VREF
160
VDD
R3
MN5
VOSC
NO
111
112
121
122
131
132
140
151
152
COM1 ("LOW")
COM2 ("HIGH")
MP1
ON
R1
C1
VOSC1
N1
MN1
OFF
MP2
OFF
R2
C2
0V
N2
MN2
ON
MP3
ON
MN3
MP4
OFF
MN4
LAT1 ("LOW")
LAT2 ("HIGH")
Q
S
OB
R
SR LATCH
EN ("HIGH")

FIG.6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100
200
210
VC
C3
R4
VREF
160
VDD
R3
MN5
VOSC
NO
111
112
121
122
131
132
COM1 ("LOW")
COM2 ("HIGH")
VDD
MP1
OFF
R1
C1
0V
N1
MN1
ON
VDD
MP2
ON
R2
C2
VOSC2
N2
MN2
OFF
MP3
OFF
MN3
MP4
ON
MN4
151
152
LAT1 ("HIGH")
LAT2 ("LOW")
140
Q
S
QB
R
SR LATCH
EN ("HIGH")

FIG.7

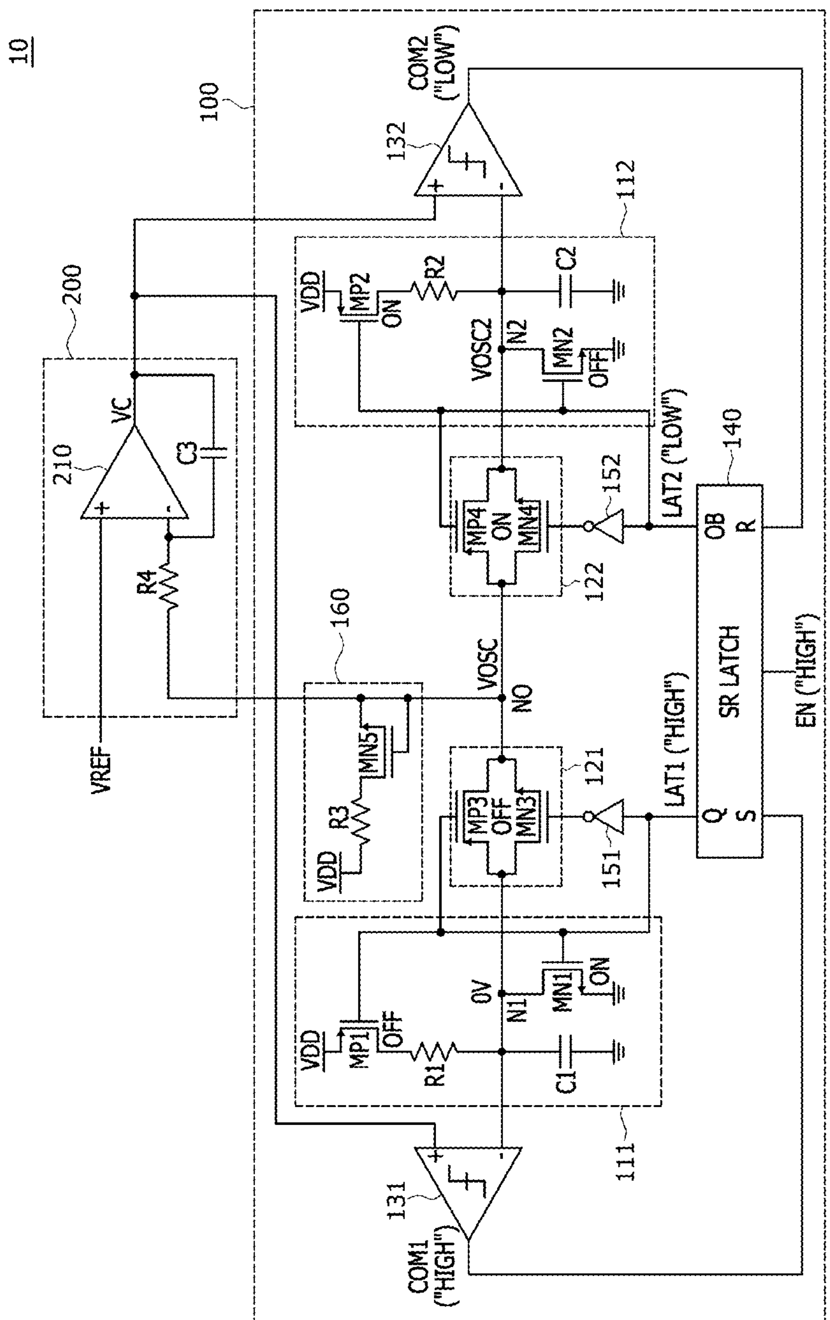

10
100
200
210
VC
C3
R4
VREF
160
VDD
R3
MN5
VOSC
NO
111
VDD
MP1
OFF
R1
C1
0V
N1
MN1
ON
112
VDD
MP2
ON
R2
C2
VOSC2
N2
MN2
OFF
121
MP3
OFF
MN3
122
MP4
ON
MN4
151
152
131
COM1
("HIGH")
132
COM2
("LOW")
140
SR LATCH
Q
S
OB
R
LAT1 ("HIGH")
LAT2 ("LOW")
EN ("HIGH")

FIG.8

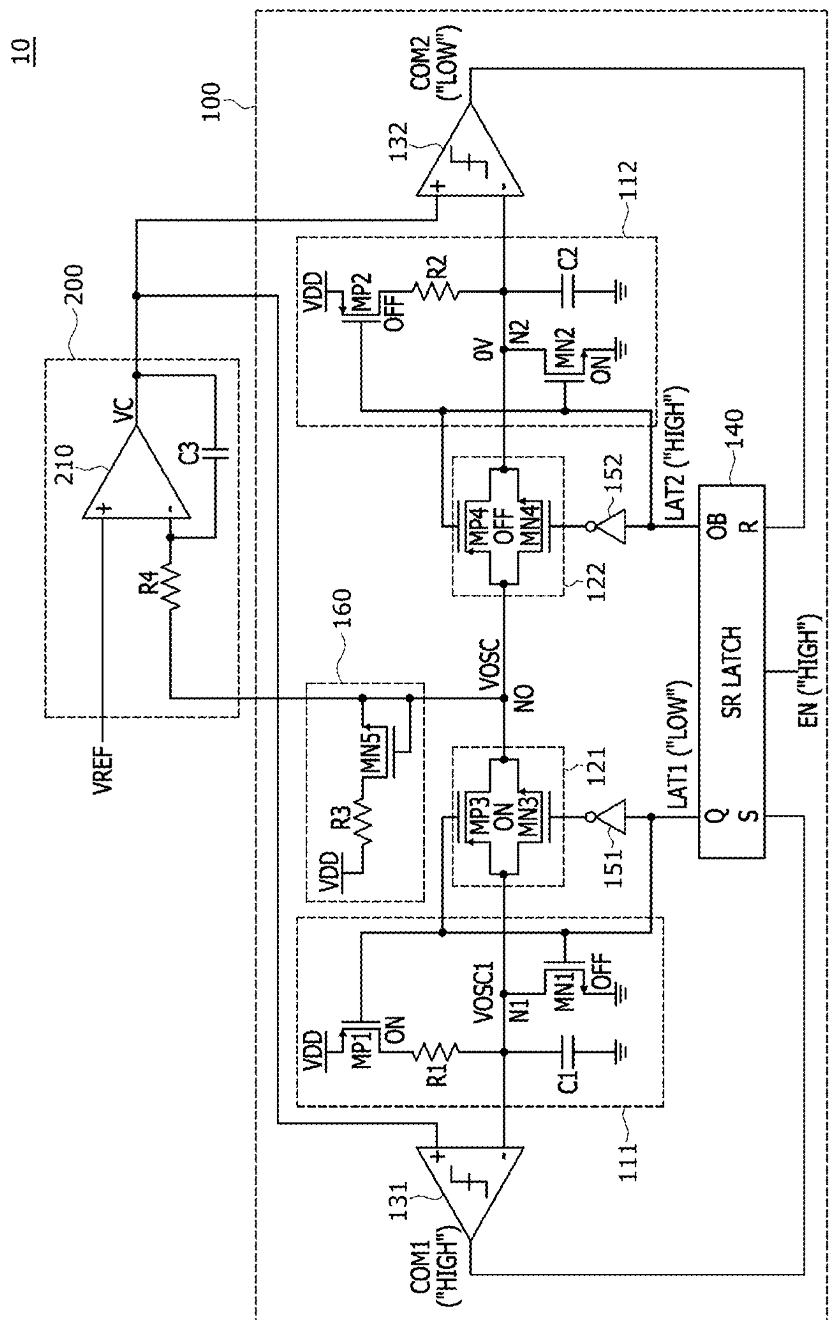

10
100
200
210
VC
C3
R4
VREF
160
VOSC
NO
R3
MN5
VDD
COM2
("LOW")
132
112
VDD
MP2
OFF
R2
C2
0V
N2
MN2
ON
122
MP4
OFF
MN4
152
LAT2 ("HIGH")
140
QB
R
SR LATCH
EN ("HIGH")
Q
S
LAT1 ("LOW")
121
MP3
ON
MN3
151
111
VDD
MP1
ON
R1
C1
VOSC1
N1
MN1
OFF
131
COM1
("HIGH")

FIG.9

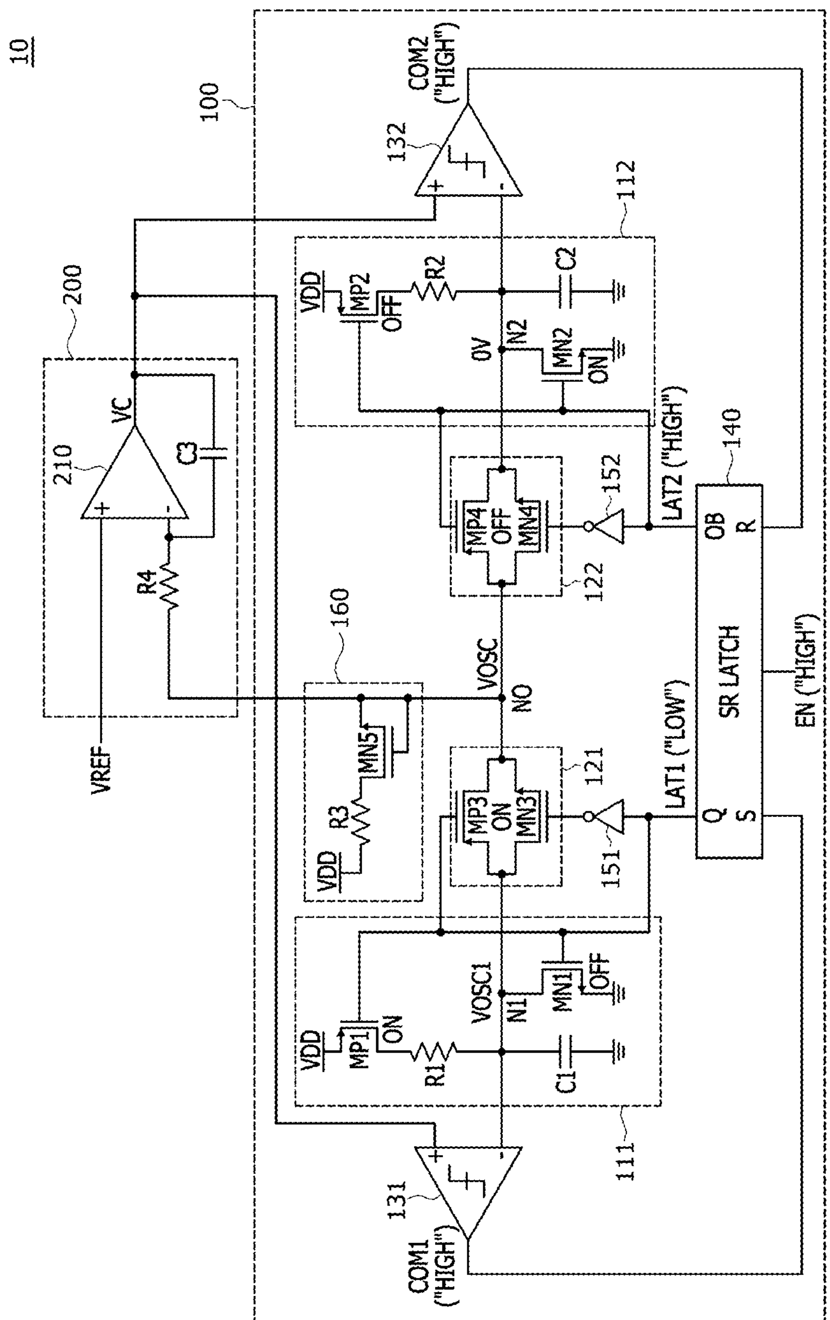

10
100
200
210
VC
C3
R4
VREF
160
VDD
R3
MN5
VOSC
NO
132
COM2 ("HIGH")
112
VDD
MP2
OFF
R2
C2
0V
N2
MN2
ON
MP4
OFF
MN4
122
152
LAT2 ("HIGH")
140
OB
R
SR LATCH
EN ("HIGH")
Q
S
LAT1 ("LOW")
151
121
MP3
ON
MN3
111
VDD
MP1
ON
R1
C1
VOSC1
N1
MN1
OFF
131
COM1 ("HIGH")

FIG.10

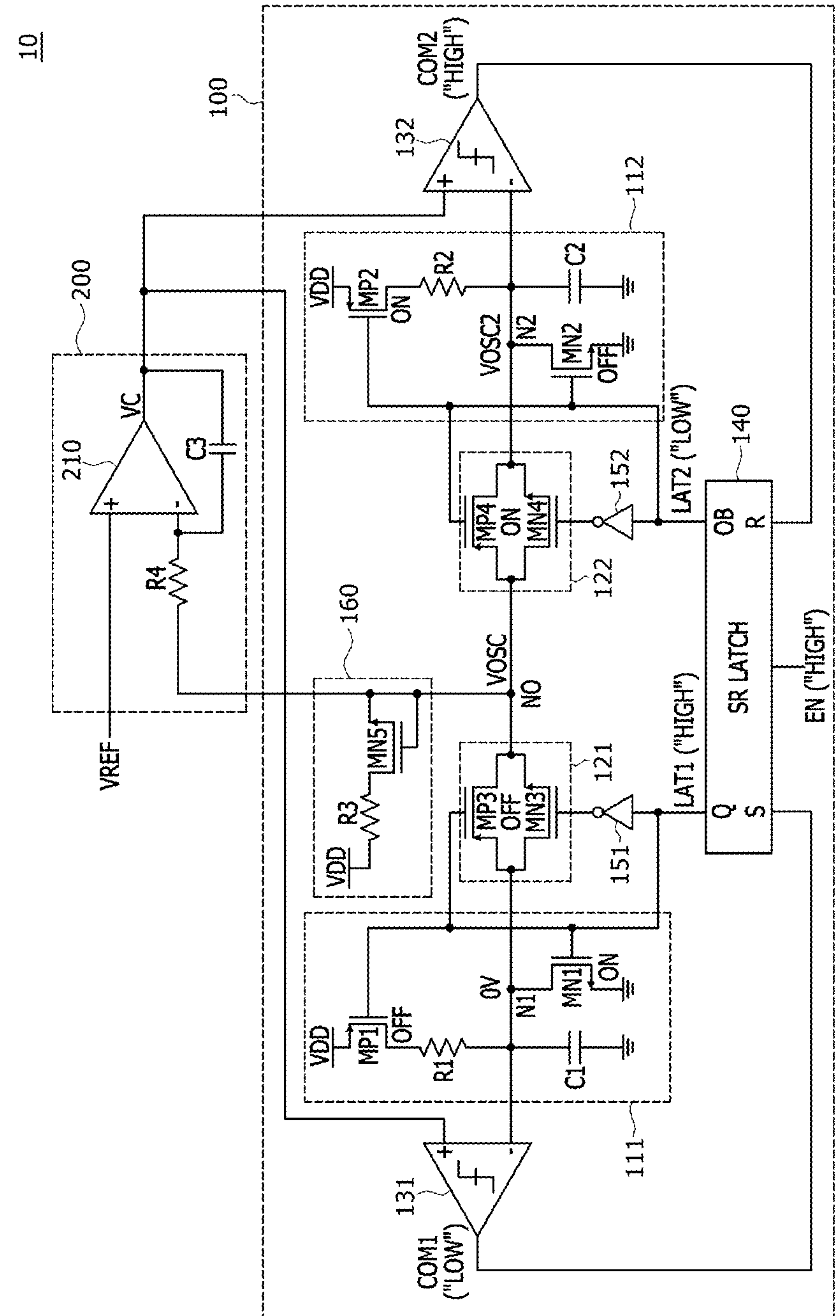

10
100
200
210
VC
C3
R4
VREF
160
VOSC
NO
VDD
R3
MN5
111
112
121
122
131
132
140
151
152
COM1 ("LOW")
COM2 ("HIGH")
VDD
MP1
OFF
R1
C1
N1
0V
MN1
ON
MP2
ON
R2
C2
VOSC2
N2
MN2
OFF
MP3
OFF
MN3
MP4
ON
MN4
LAT1 ("HIGH")
LAT2 ("LOW")
Q
S
QB
R
SR LATCH
EN ("HIGH")

FIG.11

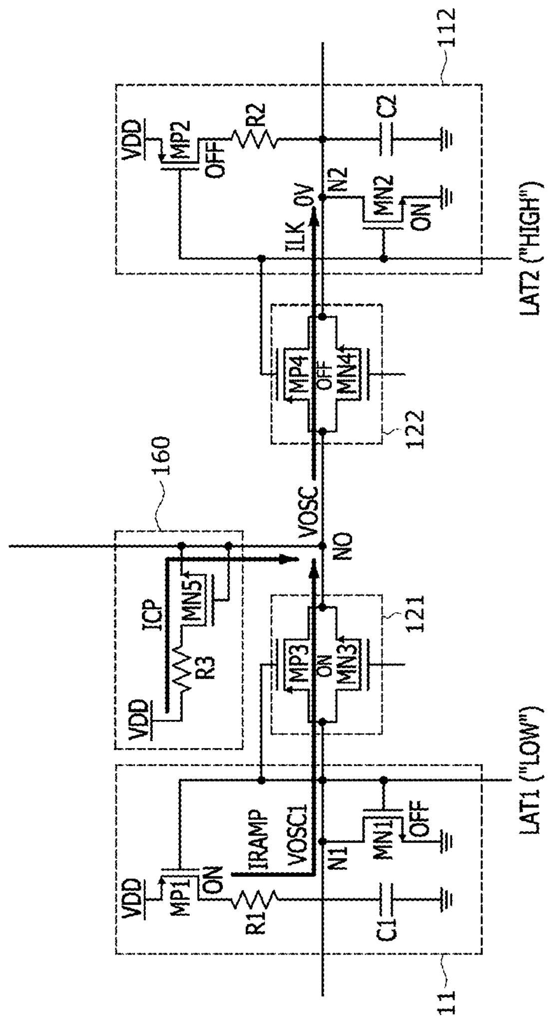

VDD
MP1
ON
R1
C1
IRAMP
VOSC1
N1
MN1
OFF
LAT1 ("LOW")
111
VDD
R3
ICP
MN5
160
MP3
ON
MN3
121
VOSC
NO
MP4
OFF
MN4
122
ILK
0V
N2
MN2
ON
VDD
MP2
OFF
R2
C2
LAT2 ("HIGH")
112

FIG.12

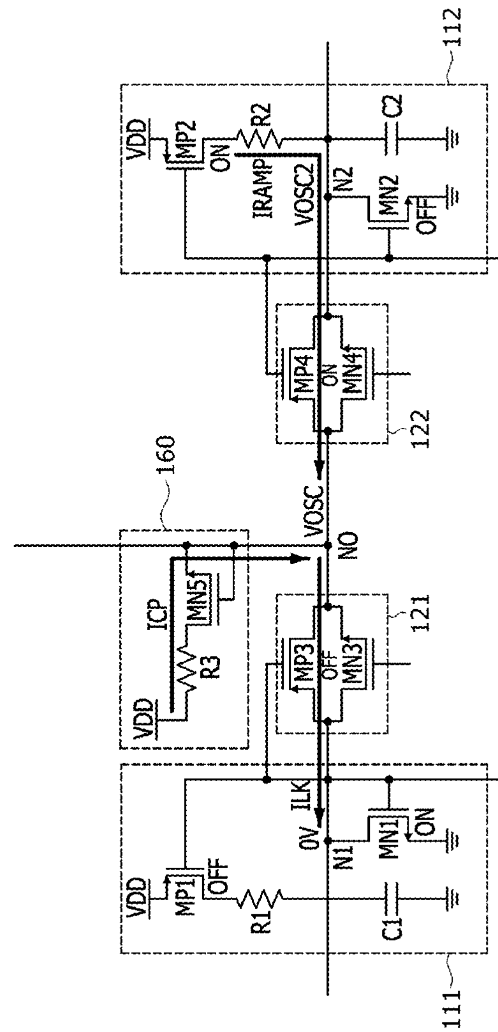

112
VDD
MP2
ON
R2
C2
IRAMP
VOSC2
N2
MN2
OFF
LAT2 ("LOW")
MP4
ON
MN4
122
160
VOSC
NO
ICP
MN5
VDD
R3
MP3
OFF
MN3
121
LAT1 ("HIGH")
VDD
MP1
OFF
R1
C1
0V
ILK
N1
MN1
ON
111

FIG.13

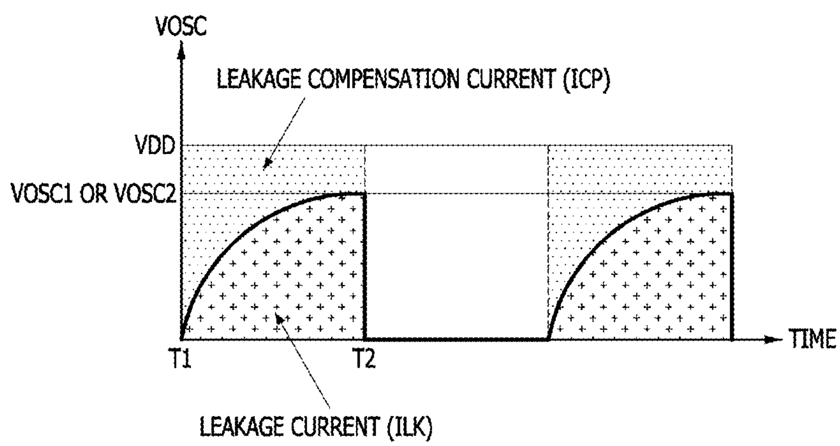

VOSC
LEAKAGE COMPENSATION CURRENT (ICP)
VDD
VOSC1 OR VOSC2
TIME
T1
T2
LEAKAGE CURRENT (ILK)

FIG.14

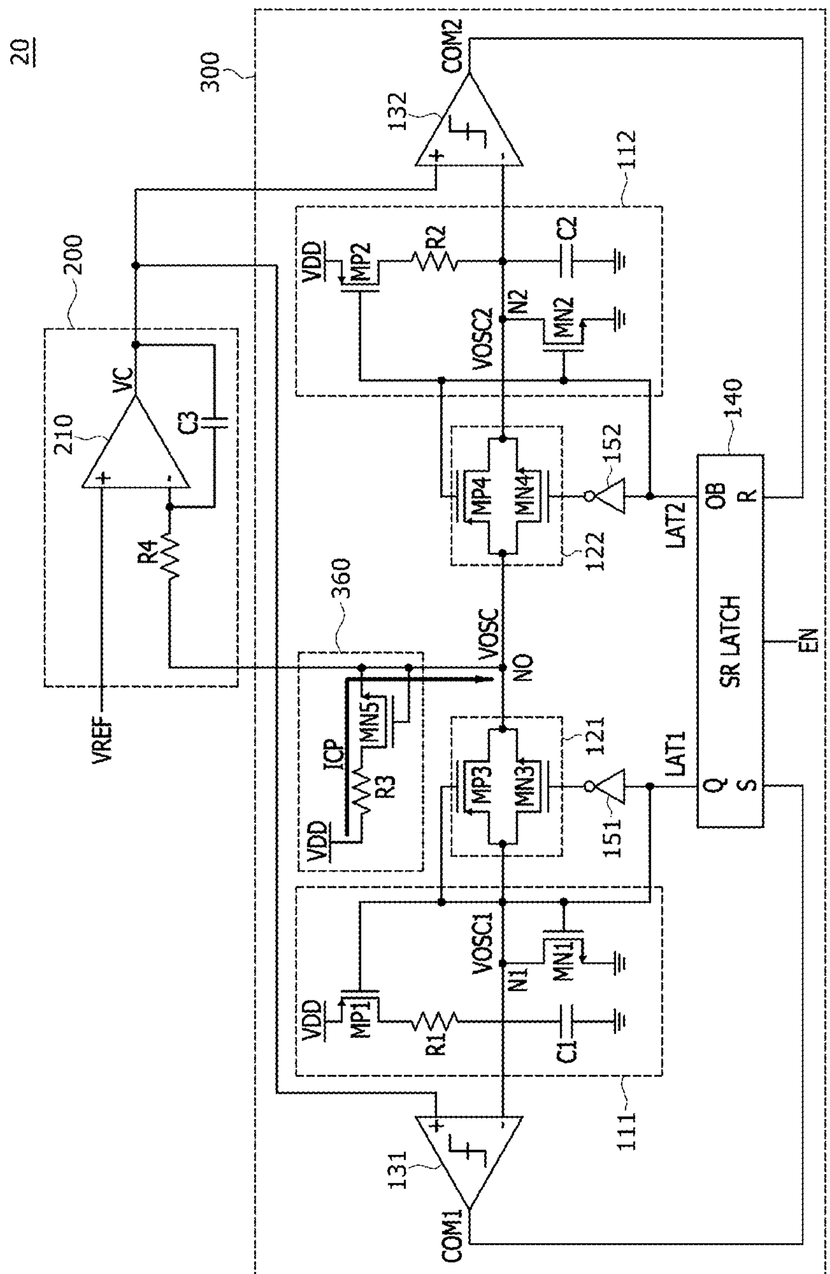

20
300
200
210
VC
C3
R4
VREF
360
ICP
R3
MN5
VDD
VOSC
NO
132
COM2
112
MP2
R2
C2
VOSC2
N2
MN2
MP4
MN4
152
122
LAT2
140
OB
R
SR LATCH
EN
Q
S
LAT1
151
121
MP3
MN3
111
MP1
R1
C1
VOSC1
N1
MN1
131
COM1

FIG.15

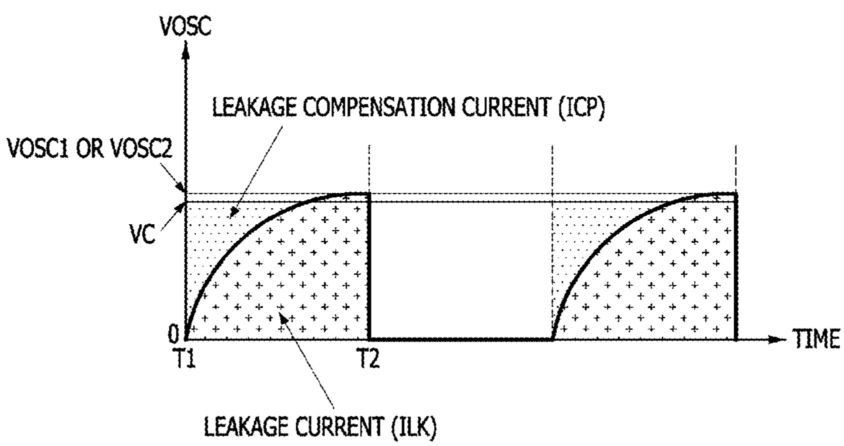

VOSC
LEAKAGE COMPENSATION CURRENT (ICP)
VOSC1 OR VOSC2
VC
0
T1
T2
TIME
LEAKAGE CURRENT (ILK)

FIG.16

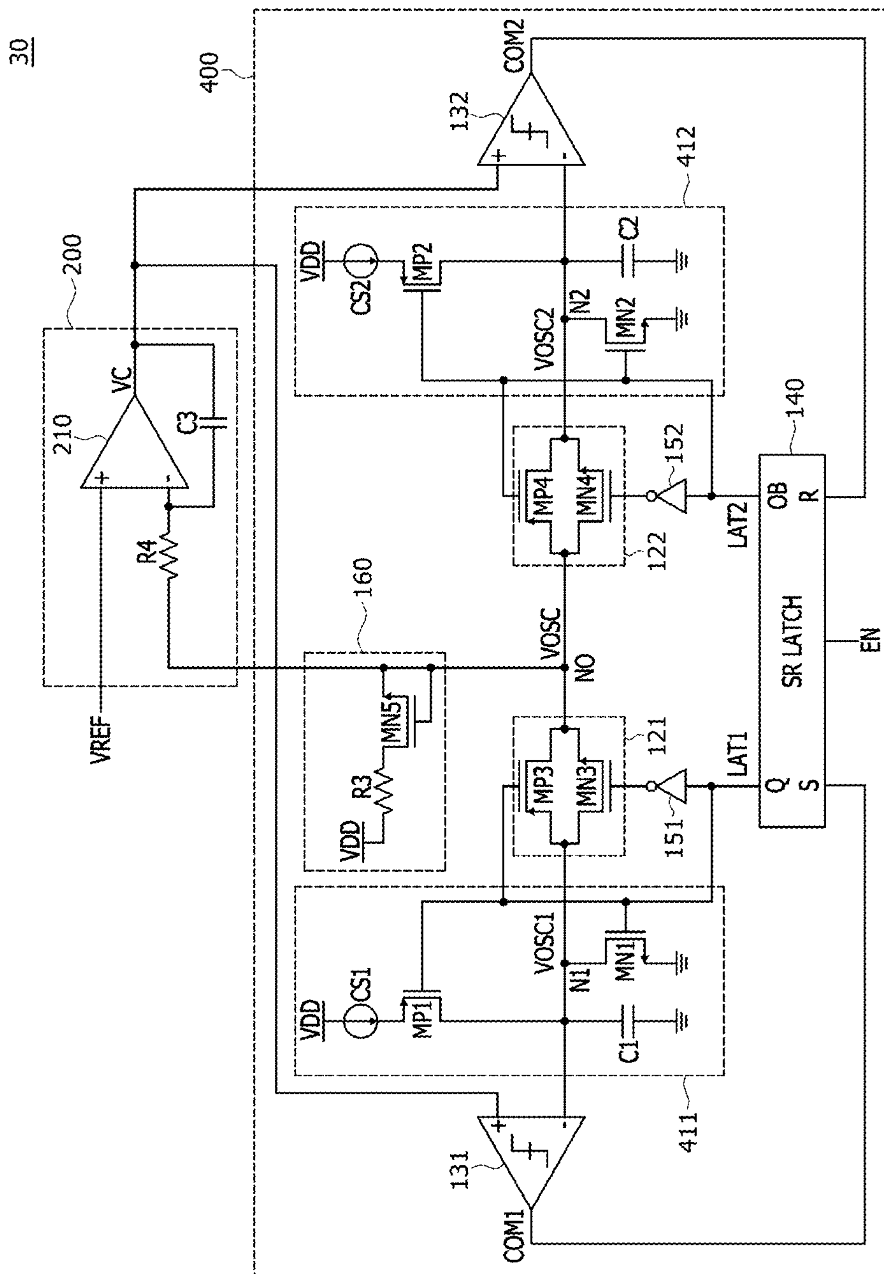

30
400
200
210
VC
C3
R4
VREF
160
R3
MN5
VDD
VOSC
NO
411
412
131
132
COM1
COM2
CS1
CS2
MP1
MP2
MN1
MN2
C1
C2
N1
N2
VOSC1
VOSC2
121
122
MP3
MN3
MP4
MN4
151
152
LAT1
LAT2
140
SR LATCH
Q
S
OB
R
EN

FIG.17

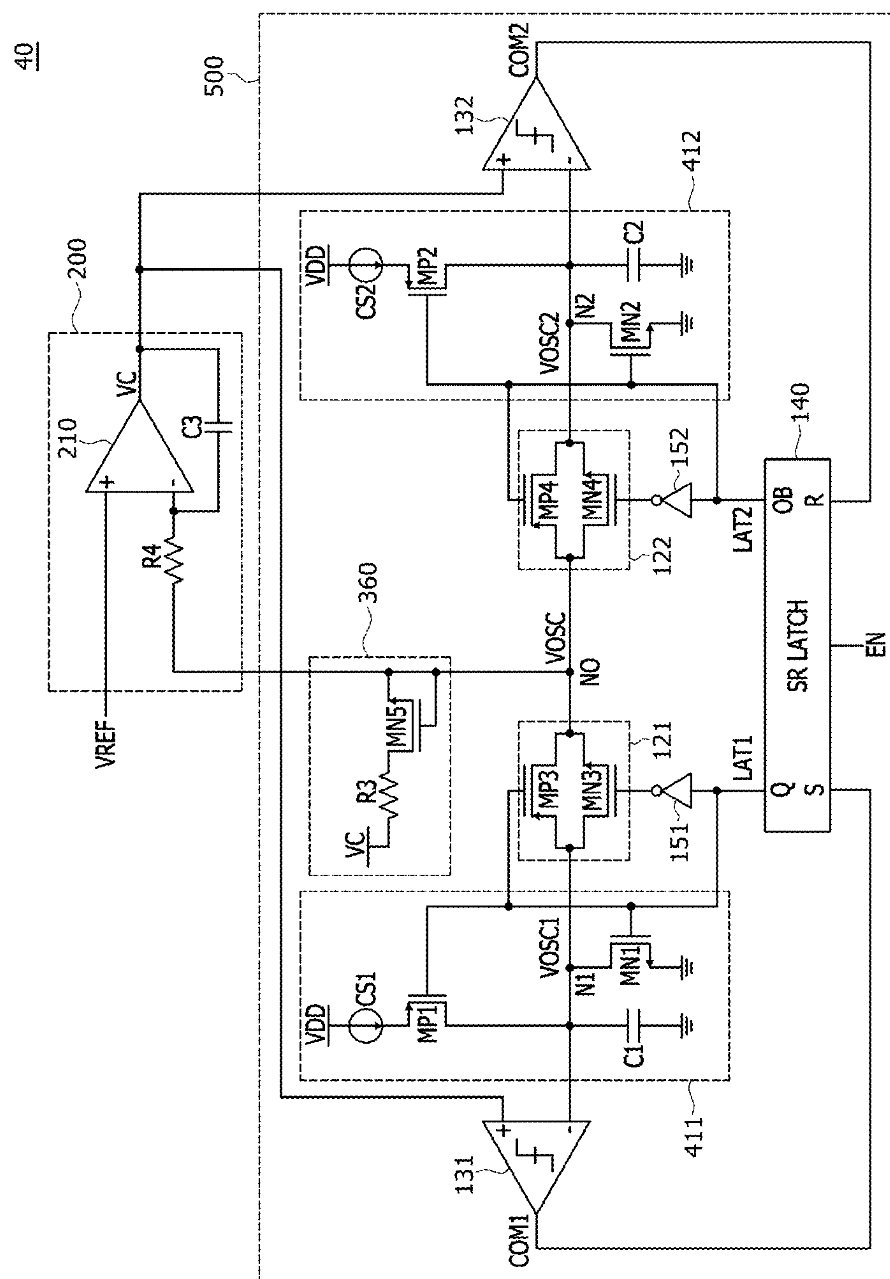
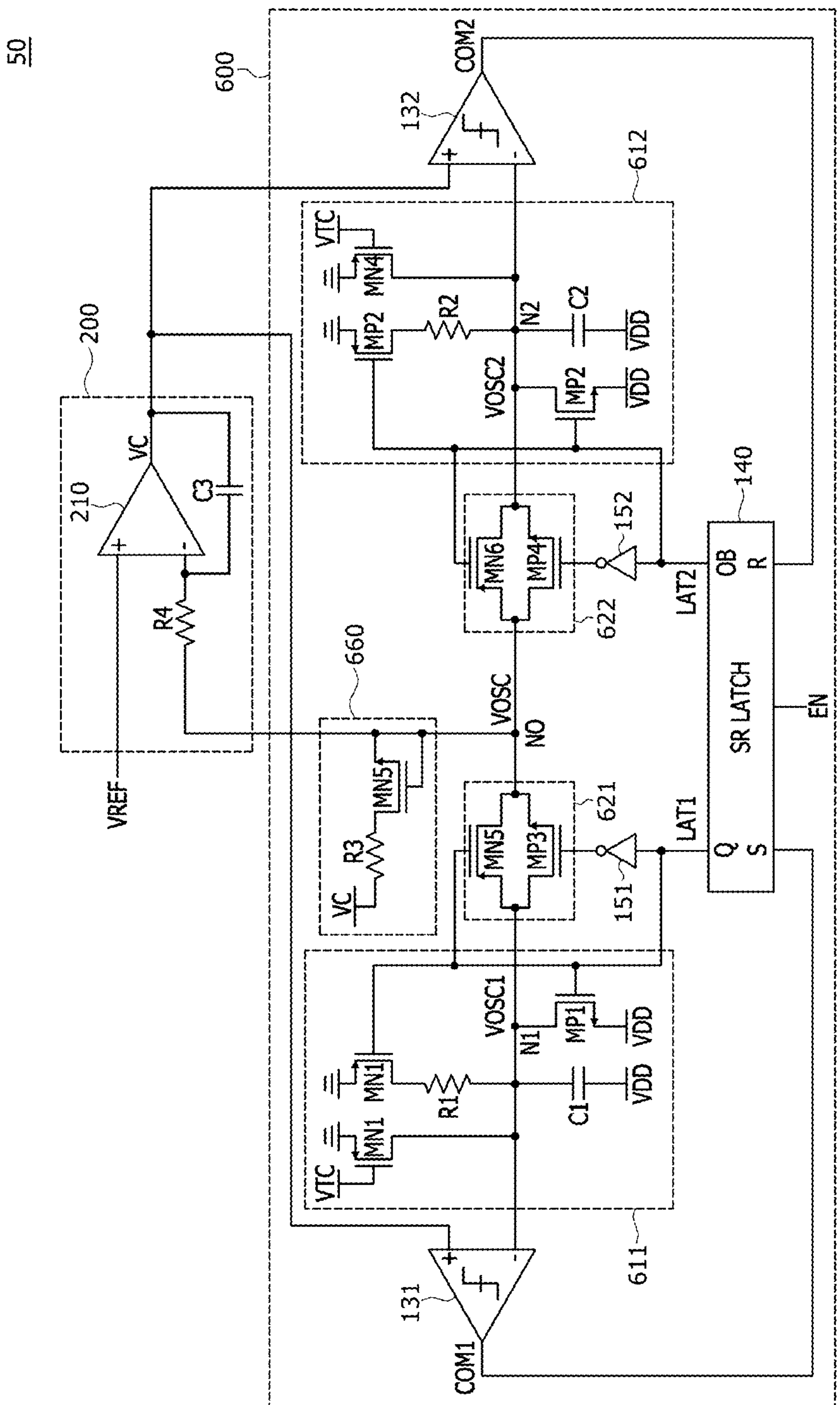

40
500
200
210
VC
C3
R4
VREF
360
R3
MN5
VC
VOSC
NO
132
COM2
412
VDD
CS2
MP2
C2
VOSC2
N2
MN2
MP4
MN4
122
152
LAT2
140
OB
R
SR LATCH
EN
Q
S
LAT1
MP3
MN3
121
151
411
VDD
CS1
MP1
C1
VOSC1
N1
MN1
131
COM1

FIG.18


50
600
200
210
VC
C3
R4
VREF
660
VC
R3
MN5
VOSC
NO
132
COM2
612
VTC
MN4
MP2
R2
N2
C2
VDD
VOSC2
MP2
VDD
622
MN6
MP4
152
LAT2
140
OB
R
SR LATCH
EN
Q
S
LAT1
621
MN5
MP3
151
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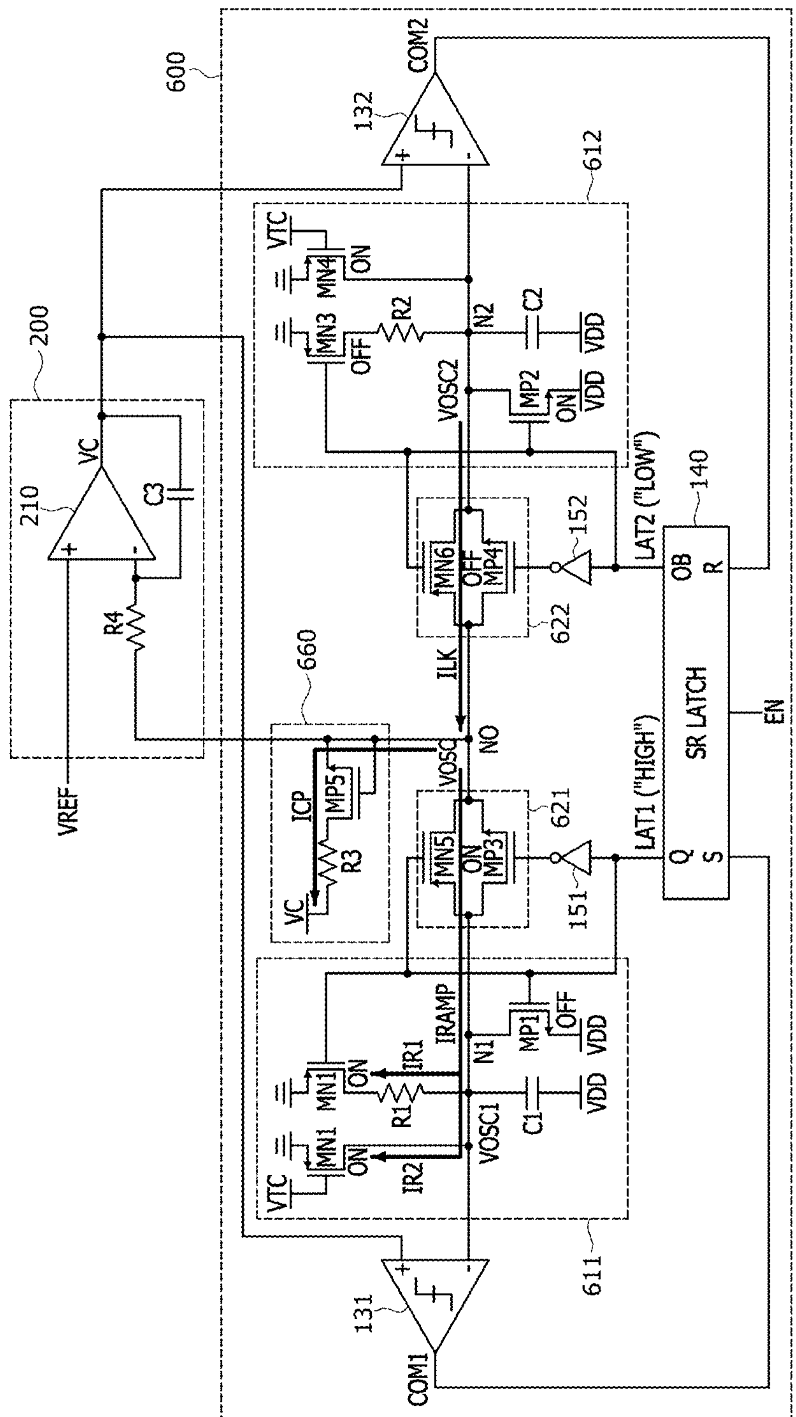
MN1
MN1
R1
C1
VDD
VOSC1
N1
MP1
VDD
131
COM1

FIG.19

50



600
200
210
VC
C3
R4
VREF
660
ICP
MP5
R3
VC
NO
VOSC
ILK
612
611
621
622
132
131
COM1
COM2
MN1
MN1
R1
IR1
IR2
IRAMP
N1
MP1
OFF
ON
C1
VDD
VOSC1
VTC
MN3
MN4
R2
N2
C2
MP2
VOSC2
MN5
MP3
MN6
MP4
151
152
LAT1 ("HIGH")
LAT2 ("LOW")
140
SR LATCH
Q
S
QB
R
EN

# OSCILLATOR WITH LEAKAGE CURRENT COMPENSATION FUNCTION

## CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2024-0030273, filed in the Korean Intellectual Property Office on Feb. 29, 2024, which is incorporated herein by reference in its entirety.

## BACKGROUND

### 1. Technical Field

[0002] Embodiments of the present disclosure generally relate to an oscillator, and more particularly, to an oscillator with a leakage current compensation function.

### 2. Related Art

[0003] In state-of-the-art technology, an entire system is integrated in a single chip. Recently, integrated circuits include analog and digital blocks required to implement high-performance and power-efficient systems. Synchronous digital circuits always require clock references. The clock reference needs to be stable over varying temperatures. External crystal oscillators achieve high-performance and are stable over a range of temperatures, but external crystal oscillators are sensitive to mechanical stress, require additional components, are expensive, require more area, and require additional pins on the chip. External components are undesirable in the recent circuit designs.

[0004] Therefore, new integrated systems require sufficiently integrated, temperature stable, accurate, and low power clock references. The frequency of the clock reference can be determined by a passive device such as a resistor, capacitor, or inductor. The on-resistance of a MOS device is not stable over a range of voltages. This is why the MOS device's influence on output frequency needs to be minimized. One problem with integrated oscillators is that the passive devices have errors depending on process and temperature, which affects the accuracy of the output frequency. Accordingly, relaxation oscillators that employ voltage averaging feedback to minimize delay in a comparator in a resistor-capacitor (RC) oscillator structure have recently been in the spotlight.

## SUMMARY

[0005] An oscillator according to an embodiment of the present disclosure may include a relaxation oscillating circuit including a first resistor-capacitor (RC) circuit generating a first oscillation voltage at a first node and a second RC circuit generating a second oscillation voltage at a second node and configured to output an oscillation voltage through an output line coupled to an output node, and a voltage averaging feedback circuit configured to receive a reference voltage and the oscillation voltage to output a control voltage through an output terminal. The relaxation oscillating circuit may include a leakage current compensation circuit coupled to the output line and configured to provide a leakage compensation current to the output node.

[0006] An oscillator according to another embodiment of the present disclosure may include a relaxation oscillating circuit including a first capacitor circuit generating a first oscillation voltage at a first node and a second capacitor circuit generating a second oscillation voltage at a second node and configured to alternately output the first oscillation voltage and the second oscillation voltage through an output line coupled to an output node, and a voltage averaging feedback circuit configured to receive a reference voltage and one of the first oscillation voltage and the second oscillation voltage and to output a control voltage through an output terminal. The relaxation oscillating circuit may include a leakage current compensation circuit coupled to the output line and configured to provide a leakage compensation current to the output node.

[0007] An oscillator according to further another embodiment of the present disclosure may include a relaxation oscillating circuit including a first resistor-capacitor (RC) circuit generating a first oscillation voltage at a first node and a second RC circuit generating a second oscillation voltage at a second node and configured to alternately output the first oscillation voltage and the second oscillation voltage through an output line coupled to an output node, and a voltage averaging feedback circuit configured to receive a reference voltage and one of the first oscillation voltage and the second oscillation voltage and to output a control voltage through an output terminal. The relaxation oscillating circuit may include a leakage current compensation circuit coupled to the output line and configured to counteract a leakage current from the output node.

## BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a circuit diagram illustrating an oscillator according to an embodiment of the present disclosure.

[0009] FIG. **2** is a circuit diagram illustrating an embodiment of a latch circuit included in an oscillator illustrated in FIG. **1**.

[0010] FIGS. **3** to **10** are circuit diagrams illustrating start-up sequence operations of an oscillator shown in FIG. **1**.

[0011] FIG. **11** is a diagram illustrating a phenomenon in which a leakage current is generated during an operation of an oscillator shown in FIG. **1** and an operation of a leakage current compensation circuit to suppress the leakage current.

[0012] FIG. **12** is a diagram illustrating a phenomenon in which a leakage current is generated during an operation of an oscillator shown in FIG. **1** and another example of an operation of a leakage current compensation circuit to suppress the leakage current.

[0013] FIG. **13** is a waveform diagram illustrating the relationship between an oscillation signal output from an output node of a relaxation oscillating circuit included in an oscillator shown in FIG. **1**, leakage current, and a leakage compensation current.

[0014] FIG. **14** is a circuit diagram illustrating an oscillator according to another embodiment of the present disclosure.

[0015] FIG. **15** is a waveform diagram illustrating a relationship between an oscillation signal output from an output node of a relaxation oscillating circuit included in an oscillator shown in FIG. **14**, a leakage current, and a leakage compensation current.

[0016] FIG. **16** is a circuit diagram illustrating an oscillator according to yet another embodiment of the present disclosure.

**[0017]** FIG. **17** is a circuit diagram illustrating an oscillator according to still another embodiment of the present disclosure.

**[0018]** FIG. **18** is a circuit diagram illustrating an oscillator according to a further embodiment of the present disclosure.

**[0019]** FIG. **19** is a circuit diagram illustrating an operation of an oscillator illustrated in FIG. **18**.

## DETAILED DESCRIPTION

**[0020]** In the following description of embodiments, it will be understood that the terms "first" and "second" are intended to identify elements, but not used to define a particular number or sequence of elements. In addition, when an element is referred to as being located "on," "over," "above," "under," or "beneath" another element, it is intended to mean relative positional relationship, but not used to limit certain cases for which the element directly contacts the other element, or at least one intervening element is present between the two elements. Accordingly, the terms such as "on," "over," "above," "under," "beneath," "below," and the like that are used herein are for the purpose of describing particular embodiments only and are not intended to limit the scope of the present disclosure.

**[0021]** Further, when an element is referred to as being "connected" or "coupled" to another element, the element may be electrically or mechanically connected or coupled to the other element directly, or may be electrically or mechanically connected or coupled to the other element indirectly with one or more additional elements between the two elements. Moreover, when a parameter is referred to as being "predetermined," it may be intended to mean that a value of the parameter is determined in advance of when the parameter is used in a process or an algorithm. The value of the parameter may be set when the process or the algorithm starts or may be set during a period in which the process or the algorithm is executed.

**[0022]** A logic "high" level and a logic "low" level may be used to describe logic levels of electric signals. A signal having a logic "high" level may be distinguished from a signal having a logic "low" level. For example, when a signal having a first voltage corresponds to a signal having a logic "high" level, a signal having a second voltage may correspond to a signal having a logic "low" level. In an embodiment, the logic "high" level may be set as a voltage level that is higher than a voltage level of the logic "low" level. Logic levels of signals may be set to be different or opposite according to embodiment. For example, a certain signal having a logic "high" level in one embodiment may be set to have a logic "low" level in another embodiment.

**[0023]** Various embodiments of the present disclosure will be described hereinafter in detail with reference to the accompanying drawings. However, the embodiments described herein are for illustrative purposes only and are not intended to limit the scope of the present disclosure.

**[0024]** FIG. **1** is a circuit diagram illustrating an oscillator according to an embodiment of the present disclosure. FIG. **2** is a circuit diagram illustrating an embodiment of a latch circuit included in an oscillator illustrated in FIG. **1**.

**[0025]** Referring to FIG. **1**, an oscillator **10** may include a relaxation oscillating circuit **100** and a voltage averaging feedback circuit **200**. The relaxation oscillating circuit **100** may include a first resistor-capacitor (RC) circuit **111**, a second RC circuit **112**, a first switching device **121**, a second switching device **122**, a first comparator **131**, a second comparator **132**, a latch circuit **140**, a first inverter **151**, a second inverter **152**, and a leakage current compensation circuit **160**.

**[0026]** The first RC circuit **111** may include a first P-channel type MOS (PMOS) transistor MP**1**, a first resistor R**1**, a first capacitor C**1**, and a first N-channel type MOS (NMOS) transistor MN**1**. A source terminal of the first PMOS transistor MP**1** may be coupled to a supply voltage VDD terminal to which a supply voltage VDD is applied. A drain terminal of the first PMOS transistor MP**1** may be coupled to a first terminal of the first resistor R**1**. A gate terminal of the first PMOS transistor MP**1** may be coupled to a first output terminal Q of the latch circuit **140**. A second terminal of the first resistor R**1** may be coupled to a first node N**1**. A first terminal and a second terminal of the first capacitor C**1** may be coupled to the first node N**1** and a ground voltage terminal to which a ground voltage is applied, respectively. A drain terminal and a source terminal of the first NMOS transistor MN**1** may be coupled to the first node N**1** and the ground voltage terminal, respectively. A gate terminal of the first NMOS transistor MN**1** may be coupled to the first output terminal Q of the latch circuit **140**.

**[0027]** When a signal of a logic "low" level is output from the first output terminal Q of the latch circuit **140**, the first PMOS transistor MP**1** may be turned on and the first NMOS transistor MN**1** may be turned off. In this case, the first capacitor C**1** may be charged, and a first oscillation voltage VOSC**1** may be applied to the first node N**1**. On the other hand, when a signal of a logic "high" level is output from the first output terminal Q of the latch circuit **140**, the first PMOS transistor MP**1** may be turned off and the first NMOS transistor MN**1** may be turned on. In this case, the first capacitor C**1** may be discharged, and the voltage of the first node N**1** may drop to the ground voltage.

**[0028]** The second RC circuit **112** may include a second PMOS transistor MP**2**, a second resistor R**2**, a second capacitor C**2**, and a second NMOS transistor MN**2**. A source terminal of the second PMOS transistor MP**2** may be coupled to the supply voltage VDD terminal. A drain terminal of the second PMOS transistor MP**2** may be coupled to a first terminal of the second resistor R**2**. A gate terminal of the second PMOS transistor MP**2** may be coupled to a second output terminal QB of the latch circuit **140**. A second terminal of the second resistor R**2** may be coupled to a second node N**2**. A first terminal and a second terminal of the second capacitor C**2** may be coupled to the second node N**2** and the ground voltage terminal, respectively. A drain terminal and a source terminal of the second NMOS transistor MN**2** may be coupled to the second node N**2** and the ground voltage terminal, respectively. A gate terminal of the second NMOS transistor MN**2** may be coupled to the second output terminal QB of the latch circuit **140**.

**[0029]** When a signal of a logic "low" level is output from the second output terminal QB of the latch circuit **140**, the second PMOS transistor MP**2** may be turned on and the second NMOS transistor MN**2** may be turned off. In this case, the second capacitor C**2** may be charged, and a second oscillation voltage VOSC**2** may be applied to the second node N**2**. On the other hand, when a signal of a logic "high" level is output from the second output terminal QB of the latch circuit **140**, the second PMOS transistor MP**2** may be turned off and the second NMOS transistor MN**2** may be turned on. In this case, the second capacitor C2 may be discharged, and the voltage of the second node N2 may drop to the ground voltage.

**[0030]** The first switching device **121** may be coupled to the first node N1 and an output node NO. The first switching device **121** may have a structure of a transmission gate including a third PMOS transistor MP3 and a third NMOS transistor MN3. In an embodiment, each of the third PMOS transistor MP3 and the third NMOS transistor MN3 may have low voltage threshold (LVT) characteristics or ultra-low voltage threshold (ULVT) characteristics. A source terminal of the third PMOS transistor MP3 and a drain terminal of the third NMOS transistor MN3 may be coupled to the first node N1. A drain terminal of the third PMOS transistor MP3 and a source terminal of the third NMOS transistor MN3 may be coupled to the output node NO. A first latch signal LAT1 output from the first output terminal Q of the latch circuit **140** may be applied to a gate terminal of the third PMOS transistor MP3. An output signal from the first inverter **151** (i.e., an inverted signal of the first latch signal LAT1 output from the first output terminal Q of the latch circuit **140**) may be applied to a gate terminal of the third NMOS transistor MN3. When a signal of a logic "low" level is output through the first output terminal Q of the latch circuit **140**, the first switching device **121** may short the first node N1 and the output node NO. On the other hand, when a signal of a logic "high" level is output through the first output terminal Q of the latch circuit **140**, the first switching device **121** may open the first node N1 and the output node NO.

**[0031]** The second switching device **122** may be coupled to the output node NO and the second node N2. The second switching device **122** may have a structure of a transmission gate including a fourth PMOS transistor MP4 and a fourth NMOS transistor MN4. In an embodiment, each of the fourth PMOS transistor MP4 and the fourth NMOS transistor MN4 may have LVT characteristics or ULVT characteristics. A source terminal of the fourth PMOS transistor MP4 and a drain terminal of the fourth NMOS transistor MN4 may be coupled to the output node NO. A drain terminal of the fourth PMOS transistor MP4 and a source terminal of the fourth NMOS transistor MN4 may be coupled to the second node N2. A second latch signal LAT2 output from the second output terminal QB of the latch circuit **140** may be applied to a gate terminal of the fourth PMOS transistor MP4. An output signal from the second inverter **152** (i.e., an inverted signal of the second latch signal LAT2 output from the second output terminal QB of the latch circuit **140**) may be applied to a gate terminal of fourth NMOS transistor MN4. When a signal of a logic "low" level is output through the second output terminal QB of the latch circuit **140**, the second switching device **122** may short the output node NO and the second node N2. On the other hand, when a signal of a logic "high" level is output through the second output terminal QB of the latch circuit **140**, the second switching device **122** may open the output node NO and the second node N2.

**[0032]** The first comparator **131** may include a first input terminal, a second input terminal, and an output terminal. The first input terminal of the first comparator **131** may be coupled to an output terminal of the voltage averaging feedback circuit **200**. The second input terminal of the first comparator **131** may be coupled to the first node N1. In an embodiment, the first input terminal and the second input terminal of the first comparator **131** may be a positive terminal and a negative terminal, respectively. The output terminal of the first comparator **131** may be coupled to a first input terminal S of the latch circuit **140**. The first comparator **131** may compare a control voltage VC output from the output terminal of the voltage averaging feedback circuit **200** with the first oscillation voltage VOSC1 at the first node N1, and output a first comparison signal COM1 having a logic level according to the comparison result. In an embodiment, when a magnitude of the first oscillation voltage VOSC1 is smaller than that of the control voltage VC, the first comparator **131** may output a signal of a logic "high" level as the first comparison signal COM1. On the other hand, when the magnitude of the first oscillation voltage VOSC1 is greater than that of the control voltage VC, the first comparator **131** may output a signal of a logic "low" level as the first comparison signal COM1.

**[0033]** The second comparator **132** may include a first input terminal, a second input terminal, and an output terminal. The first input terminal of the second comparator **132** may be coupled to the output terminal of the voltage averaging feedback circuit **200**. The second input terminal of the second comparator **132** may be coupled to the second node N2. In an embodiment, the first input terminal and the second input terminal of the second comparator **132** may be a positive terminal and a negative terminal, respectively. The output terminal of the second comparator **132** may be coupled to a second input terminal R of the latch circuit **140**. The second comparator **132** may compare the control voltage VC output from the output terminal of the voltage averaging feedback circuit **200** with a second oscillation voltage VOSC2 at the second node N2, and output a second comparison signal COM2 having a logic level according to the comparison result. In an embodiment, when a magnitude of the second oscillation voltage VOSC2 is smaller than that of the control voltage VC, the second comparator **132** may output a signal of a logic "high" level as the second comparison signal COM2. On the other hand, when the magnitude of the second oscillation voltage VOSC2 is greater than that of the control voltage VC, the second comparator **132** may output a signal of a logic "low" level as the second comparison signal COM2.

**[0034]** The latch circuit **140** may include an enable terminal to which an enable signal EN is input, the first input terminal S to which the first comparison signal COM1 from the first comparator **131** is input, and the second input terminal R to which the second comparison signal COM2 from the second comparator **132** is input. The latch circuit **140** may include the first output terminal Q and the second output terminal QB. The first output terminal Q of the latch circuit **140** may be coupled to the gate terminals of the first PMOS transistor MP1 and the first NMOS transistor MN1 included in the first RC circuit **111**, the gate terminal of the third PMOS transistor MP3 of the first switching device **121**, and the input terminal of the first inverter **151**. The second output terminal QB of the latch circuit **140** may be coupled to the gate terminals of the second PMOS transistor MP2 and the second NMOS transistor MN2 included in the second RC circuit **112**, the gate terminal of the fourth PMOS transistor MP4 of the second switching device **122**, and the input terminal of the second inverter **152**.

**[0035]** Referring to FIG. **2**, the latch circuit **140** may be a gate set-reset (SR) NAND latch circuit. The latch circuit **140** may include a first buffer **141**, a second buffer **142**, a first

NAND gate **143**, and a second NAND gate **144**. An input terminal of the first buffer **141** may be coupled to the enable terminal to which the enable signal EN is input. An output terminal of the first buffer **141** may be coupled to an input terminal of the second buffer **142**. An output terminal of the second buffer **142** may be coupled to one of three input terminals of the second NAND gate **144**. The first buffer **141** and the second buffer **142** may delay the enable signal EN that is input through the enable terminal from reaching the second NAND gate **144** by delaying a time to transmit the enable signal EN. In the process of performing the start-up sequence of the oscillator (**10** in FIG. **1**), even if the logic level of the enable signal EN transitions from a logic "low" level to a logic "high" level, the enable signal EN of the logic "high" level may be transmitted to the second NAND gate **144** when the delay time of the first buffer **141** and the second buffer **142** has elapsed. In this embodiment, the two buffers **141** and **142** are used as an example, but the number of buffers may be more than two.

**[0036]** The first NAND gate **143** may have three input terminals and one output terminal. One of the three input terminals of the first NAND gate **143** may be coupled to the enable terminal to which the enable signal EN is input. Another one of the three input terminals of the first NAND gate **143** may be coupled to the first input terminal S of the latch circuit **140**. Accordingly, the first NAND gate **143** may receive the first comparison signal COM**1** input from the first input terminal S. The remaining one of the three input terminals of the first NAND gate **143** may be coupled to an output terminal of the second NAND gate **144**, that is, the second output terminal QB of the latch circuit **140**. The output terminal of the first NAND gate **143** may be coupled to one of the input terminals of the second NAND gate **144** and the first output terminal Q of the latch circuit **140**. The first NAND gate **143** may output a first latch signal LAT**1** through the output terminal.

**[0037]** Similar to the first NAND gate **143**, the second NAND gate **144** may also have three input terminals and one output terminal. One of the three input terminals of the second NAND gate **144** may be coupled to the output terminal of the second buffer **142**. Another one of the three input terminals of the second NAND gate **144** may be coupled to the second input terminal R of the latch circuit **140**. Accordingly, the second NAND gate **144** may receive the second comparison signal COM**2** input from the second input terminal R of the latch circuit **140**. The remaining one of the three input terminals of the second NAND gate **144** may be coupled to the output terminal of the first NAND gate **143**, that is, the first output terminal Q of the latch circuit **140**. The output terminal of the second NAND gate **144** may be coupled to one of the input terminals of the first NAND gate **143** and the second output terminal QB of the latch circuit **140**. The second NAND gate **144** may output a second latch signal LAT**2** through the output terminal.

**[0038]** Referring back to FIG. **1**, the first inverter **151** of the relaxation oscillating circuit **100** may receive the first latch signal LAT**1**, which is output from the first output terminal Q of the latch circuit **140**, through an input terminal. The first inverter **151** may output an inverted signal of the first latch signal LAT**1** through an output terminal. The output terminal of the first inverter **151** may be coupled to the gate terminal of the third NMOS transistor MN**3** of the first switching device **121**. The second inverter **152** of the relaxation oscillating circuit **100** may receive the second latch signal LAT**2**, which is output from the second output terminal QB of the latch circuit **140**, through an input terminal. The second inverter **152** may output an inverted signal of the second latch signal LAT**2** through an output terminal. The output terminal of the second inverter **152** may be coupled to the gate terminal of the fourth NMOS transistor MN**4** of the second switching device **122**.

**[0039]** The leakage current compensation circuit **160** may include a third resistor R**3** and a fifth NMOS transistor MN**5**. The third resistor R**3** may be coupled to the supply voltage VDD terminal and a drain terminal of the fifth NMOS transistor MN**5**. A gate terminal and a source terminal of the fifth NMOS transistor MN**5** may each be coupled to the output node NO through an output line of the relaxation oscillating circuit **100**. In another embodiment, the leakage current compensation circuit **160** might not include the third resistor R**3**, and the drain terminal of the fifth NMOS transistor MN**5** may be directly coupled to the supply voltage VDD terminal. Because the gate terminal and source terminal of the fifth NMOS transistor MN**5** are short-circuited, a gate-source voltage VGS of the fifth NMOS transistor MN**5** may maintain 0 V. Accordingly, in the fifth NMOS transistor MN**5**, a leakage compensation current corresponding to a drain-source voltage VDS, that is, a leakage compensation current corresponding to a difference VDD-VOSC, located between the supply voltage VDD and the oscillation voltage VOSC, may flow from the drain terminal to the source terminal. Here, the leakage compensation current may be defined as the off-leakage current of the fifth NMOS transistor MN**5**. The leakage compensation current flowing from the drain terminal to the source terminal of the fifth NMOS transistor MN**5** may flow to the output node NO along the output line of the relaxation oscillating circuit **100**. The operation of the leakage current compensation circuit **160** will be described in more detail with reference to FIG. **11** below.

**[0040]** The voltage averaging feedback circuit **200** may be configured as an active filter. The voltage averaging feedback circuit **200** may include a feedback amplifier **210**, a fourth resistor R**4**, and a third capacitor C**3**. In an embodiment, the feedback amplifier **210** may be an operational amplifier. The feedback amplifier **210** may include a positive input terminal, a negative input terminal, and an output terminal. The positive input terminal of the feedback amplifier **210** may be coupled to a reference voltage VREF terminal, to which a reference voltage VREF is applied. The negative input terminal of the feedback amplifier **210** may be coupled to the fourth resistor R**4**. The output terminal of the feedback amplifier **210** may be coupled to the negative input terminal through the third capacitor C**3**. The output terminal of the feedback amplifier **210** may also be coupled to the positive input terminal of the first comparator **131** and the positive input terminal of the second comparator **132**. The fourth resistor R**4** may be coupled to the output line of the relaxation oscillating circuit **100**. The feedback amplifier **210** may output a control voltage VC through the output terminal. The control voltage VC may have a magnitude corresponding to a direct current DC component of the oscillation voltage VOSC from the output node NO, which is connected to the relaxation oscillating circuit **100** through its output line. Because the control voltage VC is transmitted to the positive input terminal of the first comparator **131** and the positive input terminal of the second comparator **132**, the influence from delay in the first comparator **131** and the second comparator **132** can be minimized.

**[0041]** FIGS. **3** to **10** are circuit diagrams illustrating start-up sequence operations of an oscillator shown in FIG. **1**. The same reference numerals as in FIG. **1** indicate the same components in FIGS. **3** to **9**, so overlapping descriptions will be omitted below.

**[0042]** Referring to FIG. **3**, when an enable signal EN input to an enable terminal EN of a latch circuit **140** is at a logic "low" level, a first latch signal LAT**1** of a logic "high" level and a second latch signal LAT**2** of a logic "high" level may be output through a first output terminal Q and a second output terminal QB of the latch circuit **140**, respectively. The first PMOS transistor MP**1** of the first RC circuit **111** may be turned off from the first latch signal LAT**1** with a logic "high" level, and the first NMOS transistor MN**1** of the first RC circuit **111** may be turned on. In addition, the third PMOS transistor MP**3** and the third NMOS transistor MN**3** of the first switching device **121** may be turned off, so that the first node N**1** and the output node NO may be opened. Accordingly, the voltage at the first node N**1** may be pulled-down to the ground voltage, that is, 0 V. Similarly, the second PMOS transistor MP**2** of the second RC circuit **112** may be turned off by the second latch signal LAT**2** of a logic "high" level, and the second NMOS transistor MN**2** of the second RC circuit **112** may be turned on. Additionally, the fourth PMOS transistor MP**4** and the fourth NMOS transistor MN**4** of the second switching device **122** may be turned off, so that the output node NO and the second node N**2** may be opened. Accordingly, the voltage at the second node N**2** may also be pulled-down to the ground voltage, that is, 0 V. The initial voltage of the control voltage VC may be set to an appropriate voltage, for example, the reference voltage VREF. Accordingly, a first comparison signal COM**1** of a logic "high" level may be output from the first comparator **131**, and a second comparison signal COM**2** of a logic "high" level may be output from the second comparator **132**.

**[0043]** Next, referring to FIG. **4**, when the logic level of the enable signal EN input to the enable terminal of the latch circuit **140** transitions from a logic "low" level to a logic "high" level, as described with reference to FIG. **2**, a signal of a logic "high" level may be input to all three input terminals of the first NAND gate (**143** in FIG. **2**). Therefore, the logic level of a first latch signal LAT**1** output from the output terminal of the first NAND gate (**143** in FIG. **2**), that is, the first output terminal Q of the latch circuit **140**, may transition from a logic "high" level to a logic "low" level. On the other hand, due to the delay time of each of the buffers (**141** and **142** in FIG. **2**), an enable signal EN of a logic "low" level may be transmitted to the second NAND gate (**144** in FIG. **2**) of the latch circuit **140**. Accordingly, the logic level of the second latch signal LAT**2** output from the output terminal of the second NAND gate (**144** in FIG. **2**), that is, the second output terminal QB of the latch circuit **140**, may maintain a logic "high" level. Referring back to FIG. **4**, the third PMOS transistor MP**3** and the third NMOS transistor MN**3** of the first switching device **121** may be turned on by the first latch signal LAT**1** of a logic "low" level from first output terminal Q. On the other hand, the fourth PMOS transistor MP**4** and the fourth NMOS transistor MN**4** of the second switching device **122** may be turned off by the second latch signal LAT**2** of a logic "high" level from second output terminal QB.

**[0044]** The first RC circuit **111** may perform a first charging process with a first latch signal LAT**1** of a logic "low" level. Specifically, the first PMOS transistor MP**1** of the first RC circuit **111** may be turned on, and the first NMOS transistor MN**1** may be turned off. As the first PMOS transistor MP**1** and the first NMOS transistor MN**1** of the first RC circuit **111** are turned on and off, respectively, the first capacitor C**1** of the first RC circuit **111** may begin to be charged, and the voltage at the first node N**1** may be increased to the first oscillation voltage VOSC**1**. Because the first switching device **121** is turned on and the second switching device **122** is turned off, the oscillation voltage VOSC at the output node NO may be the same as the first oscillation voltage VOSC**1** at the first node N**1**. That is, the first oscillation voltage VOSC**1** may be transmitted from the output node NO to the voltage averaging feedback circuit **200** through the output line of the relaxation oscillating circuit **100**. In FIG. **4**, the second RC circuit **112** maintains the same state as described above with reference to FIG. **3**.

**[0045]** The output signal of the voltage averaging feedback circuit **200** may fluctuate up and down until the oscillation operation is stabilized and reaches equilibrium. Hereinafter, it is assumed that the control voltage VC output from the voltage averaging feedback circuit **200** has a magnitude that allows the direct current (DC) voltage of the oscillation voltage VOSC to be equal to the reference voltage VREF.

**[0046]** Next, referring to FIG. **5**, the control voltage VC from the voltage averaging feedback circuit **200** is input to the positive input terminal of the first comparator **131** and the positive input terminal of the second comparator **132**. When the magnitude of the first oscillation voltage VOSC**1** becomes greater than that of the control voltage VC, the logic level of the first comparison signal COM**1** output through the output terminal of the first comparator **131** may transition from a logic "high" level to a logic "low" level. The second comparison signal COM**2** output through the output terminal of the second comparator **132** may be maintained at a logic "high" level.

**[0047]** Next, referring to FIG. **6**, the first comparison signal COM**1** of a logic "low" level is transmitted from the first comparator **131** to the first input terminal S of the latch circuit **140**. As described above and with reference to FIG. **2**, the first NAND gate **143** of the latch circuit **140** may output a signal of a logic "high" level. That is, the first latch signal LAT**1** of a logic "high" level may be output through the first output terminal Q of the latch circuit **140**. With the first NAND gate **143** output of a signal of a logic "high" level and the enable signal EN of a logic "high" level input to the second NAND gate **144**, signal of a logic "high" level may be input to all three input terminals of the second NAND gate **144** of the latch circuit **140** illustrated in FIG. **2**. Accordingly, the second NAND gate **144** of the latch circuit **140** may output a signal of a logic "low" level. That is, the second latch signal LAT**2** of a logic "low" level may be output through the second output terminal QB of the latch circuit **140**. The third PMOS transistor MP**3** and the third NMOS transistor MN**3** of the first switching device **121** may be turned off by the first latch signal LAT**1** of a logic "high" level. In addition, the fourth PMOS transistor MP**4** and the fourth NMOS transistor MN**4** of the second switching device **122** may be turned on by the second latch signal LAT**2** of a logic "low" level.

**[0048]** With the first latch signal LAT**1** at a logic "high" level, the first RC circuit **111** may perform a first discharging process. Specifically, the first PMOS transistor MP**1** of the first RC circuit **111** may be turned off, and the first NMOS transistor MN**1** may be turned on. The first switching device **121** is turned off and first node N**1** and the output node NO are opened, and the first PMOS transistor MP**1** and the first NMOS transistor MN**1** of the first RC circuit **111** are turned off and turned on, respectively. As a result, the first capacitor C**1** of the first RC circuit **111** may begin to be discharged and the voltage at the first node N**1** may drop to 0 V.

**[0049]** On the other hand, with the second latch signal LAT**2** at a logic "low" level, the second RC circuit **112** may perform a first charging process. Specifically, the second PMOS transistor MP**2** of the second RC circuit **112** may be turned on, and the second NMOS transistor MN**2** of the second RC circuit **112** may be turned off. Accordingly, the second capacitor C**2** of the second RC circuit **112** may begin to be charged, and the voltage at the second node N**2** may be increased to the second oscillation voltage VOSC**2**. Because the first switching device **121** is turned off and the second switching device **122** is turned on, the oscillation voltage VOSC at the output node NO may be equal to the second oscillation voltage VOSC**2** at the second node N**2**. That is, the second oscillation voltage VOSC**2** may be transmitted from the output node NO to the voltage averaging feedback circuit **200** through the output line of the relaxation oscillating circuit **100**.

**[0050]** Next, referring to FIG. **7**, the voltage at the first node N**1** may be pulled down to 0 V from the first discharging process of the first RC circuit **111**. Accordingly, the logic level of the first comparison signal COM**1** output from the output terminal of the first comparator **131** may transition from a logic "low" level to a logic "high" level. The first comparison signal COM**1** of a logic "high" level is input to the first input terminal S of the latch circuit **140**. Even though a signal of a logic "low" level output from the second NAND gate **144** is input to one of the three input terminals of the first NAND gate **143** of the latch circuit **140**, the first latch signal LAT**1** output from the first output terminal Q of the latch circuit **140** remains at a logic "high" level. When the magnitude of the second oscillation voltage VOSC**2** becomes greater than that of the control voltage VC during the first charging process in the second RC circuit **112**, the logic level of the second comparison signal COM**2** output through the output terminal of the second comparator **132** may transition from a logic "high" level to a logic "low" level.

**[0051]** Next, referring to FIG. **8**, as the second comparison signal COM**2** of a logic "low" level output from the second comparator **132** is input to the second input terminal R of the latch circuit **140**, as described with reference to FIG. **2**, the second NAND gate **144** of the latch circuit **140** may output the second latch signal LAT**2** of a logic "high" level. With the second latch signal LAT**2** output at a logic "high" level, signals of a logic "high" level are input to all three input terminals of the first NAND gate **143** of the latch circuit **140**. As a result, the first NAND gate **143** of the latch circuit **140** may output the first latch signal LAT**1** of a logic "low" level The third PMOS transistor MP**3** and the third NMOS transistor MN**3** of the first switching device **121** may be turned on by the first latch signal LAT**1** of a logic "low" level. In addition, the fourth PMOS transistor MP**4** and the fourth NMOS transistor MN**4** of the second switching device **122** may be turned off by the second latch signal LAT**2** of a logic "high" level.

**[0052]** With the first latch signal LAT**1** at a logic "low" level, the first RC circuit **111** may perform a second charging process. Specifically, the first PMOS transistor MP**1** of the first RC circuit **111** may be turned on, and the first NMOS transistor MN**1** of the first RC circuit **111** may be turned off. Accordingly, the first capacitor C**1** of the first RC circuit **111** may begin to be charged, and the voltage at the first node N**1** may be increased to the first oscillation voltage VOSC**1**. Because the first switching device **121** is in a turned-on state and the second switching device **122** is in a turned-off state, the oscillation voltage VOSC at the output node NO may be equal to the first oscillation voltage VOSC**1** at the first node N**1**. That is, the first oscillation voltage VOSC**1** may be transmitted from the output node NO to the voltage averaging feedback circuit **200** through the output line of the relaxation oscillating circuit **100**.

**[0053]** On the other hand, with the second latch signal LAT**2** at a logic "high" level, the second RC circuit **112** may perform a first discharging process. Specifically, the second PMOS transistor MP**2** of the second RC circuit **112** may be turned off, and the second NMOS transistor MN**2** of the second RC circuit **112** may be turned on. The second switching device **122** is turned off so that the output node NO and the first node N**1** are open-circuited, and the second PMOS transistor MP**2** and the second NMOS transistor MN**2** of the second RC circuit **112** are turned off and turned on, respectively. The second capacitor C**2** of the second RC circuit **112** may begin to be discharged, and the voltage at the second node N**2** may drop to 0 V.

**[0054]** Next, referring to FIG. **9**, because the voltage at the second node N**2** is pulled-down through the first discharging process of the second RC circuit **112**, the logic level of the second comparison signal COM**2** output from the output terminal of the second comparator **132** may transition from a logic "low" level to a logic "high" level. As described above with reference to FIG. **2**, the second comparison signal COM**2** of a logic "high" level is input to the second input terminal R of the latch circuit **140**, and a signal of a logic "low" level output from the first NAND gate **143** is input to one of the three input terminals of the second NAND gate **144** of the latch circuit **140**. As a result, the second latch signal LAT**2** output from the second output terminal QB of the latch circuit **140** may be maintained at a logic "high" level.

**[0055]** Next, referring to FIG. **10**, when the magnitude of the first oscillation voltage VOSC**1** becomes greater than that of the control voltage VC in the second discharging process of the first RC circuit **111**, the logic level of the first comparison signal COM**1** output through the output terminal of the first comparator **131** may transition from a logic "high" level to a logic "low" level. The latch circuit **140** that receives the first comparison signal COM**1** of a logic "low" level through the first input terminal S may output the first latch signal LAT**1** of a logic "high" level through the first output terminal Q and output the second latch signal LAT**2** of a logic "low" level through the second output terminal QB. The third PMOS transistor MP**3** and the third NMOS transistor MN**3** of the first switching device **121** may be turned off by the first latch signal LAT**1** of a logic "high" level. The third PMOS transistor MP**3** and the third NMOS transistor MN**3** of the second switching device **122** may be turned on by the second latch signal LAT**2** of a logic "low" level. With the first latch signal LAT1 at a logic "high" level, the first RC circuit **111** may perform a second discharging process. On the other hand, with the second latch signal LAT2 at a logic "low" level, the second RC circuit **112** may perform a second charging process. The second discharging process in the first RC circuit **111** and the second charging process in the second RC circuit **122** may be performed in the same manner as the first discharging process in the first RC circuit **111** and the first charging process in the second RC circuit **112** described above and with reference to FIG. **5**, respectively, and so overlapping descriptions will be omitted.

**[0056]** When the second discharging process in the first RC circuit **111** and the second charging process in the second RC circuit **112** are performed, a third charging process in the first RC circuit **111** and a second discharging process in the second RC circuit **112** may be performed in the same manner as the second charging process in the first RC circuit **111** and the first discharging process in the second RC circuit **112** described above and with reference to FIG. **7**, respectively. Charging the first RC circuit **111** and discharging the second RC circuit **112**, and discharging the first RC circuit **111** and charging the second RC circuit **112** are alternately performed. Consequently, oscillator **10** may be stabilized and a start-up sequence of the oscillator **10** may be completed.

**[0057]** After the oscillator **10** is stabilized by alternately charging the first RC circuit **111** and discharging the second RC circuit **112**, then discharging the first RC circuit **111** and charging the second RC circuit **112**, the oscillation voltage VOSC, which is the sum of the first oscillation voltage VOSC1 at the first node N1 and the second oscillation voltage VOSC2 at the second node N2, may be output from the output node NO of the relaxation oscillating circuit **100** through its output line. The charging in the first RC circuit **111** and discharging in the second RC circuit **112** may be performed in the same manner as the second charging in the first RC circuit **111** and the first discharging in the second RC circuit **112** described above and with reference to FIG. **7**. The discharging in the first RC circuit **111** and charging in the second RC circuit **112** may be performed in the same manner as the second discharging in the first RC circuit **111** and second charging in the second RC circuit **112** described above and with reference to FIG. **10**.

**[0058]** FIG. **11** illustrates a phenomenon in which a leakage current is generated during an operation of an oscillator shown in FIG. **1** and an operation of a leakage current compensation circuit to suppress the leakage current. In FIG. **11**, among the components of an oscillator **10** described above and with reference to FIG. **1**, illustration of components are omitted except for the first RC circuit **111**, the second RC circuit **112**, the first switching device **121**, the second switching device **122**, and the leakage current compensation circuit **160**. In this example, it is assumed that a second latch signal LAT2 of a logic "high" level applied to the gate terminal of the fourth PMOS transistor MP4 of the second switching device **122** has the same magnitude as the supply voltage VDD.

**[0059]** Referring to FIG. **11**, during the first RC circuit **111** charging process and the second RC circuit **112** discharging process, the third PMOS transistor MP3 and the third NMOS transistor MN3 of the first switching device **121** may be turned on, and the fourth PMOS transistor MP4 and the fourth NMOS transistor MN4 of the second switching device **122** may be turned off. Accordingly, a ramp current IRAMP from the supply voltage VDD terminal may flow to the output node NO through the first PMOS transistor MP1, the first resistor R1, and the first switching device **121**. However, a portion of the ramp current IRAMP (hereinafter, referred to as "leakage current") may flow to the ground terminal through the second NMOS transistor MN2 of the second RC circuit **112** in the form of an off leakage current of the second switching device **122**.

**[0060]** For the fourth PMOS transistor MP4 of the second switching device **122**, a gate-source voltage of VOSC-VDD may be applied between the gate and source. Because the magnitude of the supply voltage VDD is greater than the magnitude of the oscillation voltage VOSC, a reverse bias may be applied between the gate terminal and the source terminal of the fourth PMOS transistor MP4. Accordingly, the leakage current through the fourth PMOS transistor MP4 may be very small, such as for example, at the level of several hundred pA. On the other hand, for the fourth NMOS transistor MN4 of the second switching device **122**, a gate-source voltage of 0 V may be applied between the gate and source. In this case, the leakage current through the fourth NMOS transistor MN4 may correspond to the magnitude of the voltage between the drain terminal and the source terminal of the fourth NMOS transistor MN4. Because the drain-source voltage of the fourth NMOS transistor MN4 is the oscillation voltage VOSC, a large amount of leakage current may flow through the fourth NMOS transistor MN4. In particular, the amount of the leakage current in the second switching device **122** may be further increased when the fourth PMOS transistor MP4 and the fourth NMOS transistor MN4 constituting the second switching device **122** are configured to have LVT characteristics or ULVT characteristics for a fast switching operation.

**[0061]** In this way, in the process of charging the first RC circuit **111** and discharging the second RC circuit **112**, as a large amount of leakage current ILK flows from the output node NO to the ground voltage terminal through the second switching device **122** in a turned-off state, the amount of the ramp current IRAMP in the first RC circuit **111** that generates the oscillation voltage VOSC at the output node NO may be decreased by the amount of the leakage current ILK. When the amount of the ramp current IRAMP is decreased, the time it takes for the voltage at the first node N1 of the first RC circuit **111** to be increased to the first oscillation voltage VOSC1 may be increased. That is, due to the leakage current ILK in the second switching device **122**, the frequency of the oscillation voltage VOSC output from the oscillator **10** may become smaller (that is, the cycle may become longer).

**[0062]** The leakage current compensation circuit **160** of the relaxation oscillating circuit **100** may supply a leakage compensation current ICP to the output node NO, thereby preventing a frequency decrease due to the leakage current ILK flowing out from the output node NO through the second switching device **122**. Specifically, because the gate terminal and source terminal of the fifth NMOS transistor MN5 of the leakage current compensation circuit **160** are short-circuited, the gate-source voltage of the fifth NMOS transistor MN5 may always be maintained at 0 V. Accordingly, the fifth NMOS transistor MN5 may be maintained in a turned-off state. That is, only the leakage compensation current ICP, which is the off leakage current, may flow from the drain terminal of the fifth NMOS transistor MN5 to the source terminal. The leakage compensation current ICP may have an amount corresponding to the drain-source voltage of the fifth NMOS transistor MN**5**, that is, the difference VDD-VOSC between the supply voltage VDD and the oscillation voltage VOSC. In an example, by appropriately adjusting the ratio (channel width/channel length) of the fifth NMOS transistor MN**5** and fourth NMOS transistor MN**4**, the leakage compensation current ICP may have the same amount of current as the leakage current ILK flowing through the second switching device **122**. The leakage compensation current ICP supplied to the output node NO from the leakage current compensation circuit **160** through the output line compensates for the decrease in the ramp current IRAMP due to the leakage current ILK at the output node NO.

**[0063]** FIG. **12** illustrates a phenomenon in which a leakage current is generated during an operation of an oscillator shown in FIG. **1** and another example of an operation of a leakage current compensation circuit to suppress the leakage current. FIG. **12** omits the components of an oscillator **10** described with reference to FIG. **1**, except for the first RC circuit **111**, the second RC circuit **112**, the first switching device **121**, the second switching device **122**, and the leakage current compensation circuit **160**. In FIG. **12**, the first latch signal LAT**1** of a logic "high" level applied to the gate terminal of the third PMOS transistor MP**3** of the first switching device **121** has the same magnitude as the supply voltage VDD. The drain and source of the MOS transistor are complementary, and accordingly, in FIG. **12**, the drain terminals and the source terminals of the third PMOS transistor MP**3** and the third NMOS transistor MN**3** are shown as source terminals and drain terminals, respectively.

**[0064]** Referring to FIG. **12**, the first RC circuit **111** performs a discharging process, the second RC circuit **112** performs a charging process, the third PMOS transistor MP**3** and the third NMOS transistor MN**3** of the first switching device **121** may be turned off, and the fourth PMOS transistor MP**4** and the fourth NMOS transistor MN**4** of the second switching device **122** may be turned on. Accordingly, the ramp current IRAMP from the supply voltage VDD terminal may flow to the output node NO through the second PMOS transistor MP**2**, the second resistor R**2**, and the second switching device **122**. However, in the present example, the leakage current corresponding to a portion of the ramp current IRAMP may flow to the ground terminal through the first NMOS transistor MN**1** of the first RC circuit **111** in the form of an off leakage current of the first switching device **121**.

**[0065]** For the third PMOS transistor MP**3** of the first switching device **121**, a gate-source voltage of VOSC-VDD may be applied between the gate and source. Because the magnitude of the supply voltage VDD is greater than that of the oscillation voltage VOSC, a reverse bias may be applied between the gate terminal and the source terminal of the third PMOS transistor MP**3**. Accordingly, the leakage current through the third PMOS transistor MP**3** may be very small, for example, at the level of several hundred pA. For the third NMOS transistor MN**3** of the first switching device **121**, a gate-source voltage of 0 V may be applied between the gate and source. In this case, the leakage current through the third NMOS transistor MN**3** may correspond to the magnitude of voltage between the drain terminal and the source terminal of the third NMOS transistor MN**3**. Because the drain-source voltage of the third NMOS transistor MN**3** is the oscillation voltage VOSC, a large amount of leakage current may flow through the third NMOS transistor MN**3**. In particular, the amount of the leakage current in the first switching device **121** may be further increased when the third PMOS transistor MP**3** and the third NMOS transistor MN**3** constituting the first switching device **121** are configured to have the LVT characteristics or the ULVT characteristics for a fast switching operation.

**[0066]** In the process of discharging the first RC circuit **111** and charging the second RC circuit **112**, a large amount of leakage current ILK flows from the output node NO to the ground voltage terminal through the first switching device **121** in a turned-off state. The amount of the ramp current IRAMP in the second RC circuit **112** that generates the oscillation voltage VOSC at the output node NO may be decreased by the amount of the leakage current ILK. When the amount of the ramp current IRAMP is decreased, the time it takes for the voltage at the second node N**2** of the second RC circuit **112** to be increased to the second oscillation voltage VOSC**2** may be increased. That is, due to the leakage current ILK in the first switching device **121**, the frequency of the oscillation voltage VOSC output from the oscillator **10** may become smaller (that is, the cycle may become longer).

**[0067]** The leakage current compensation circuit **160** may supply the leakage compensation current ICP to the output node NO, thereby preventing the frequency decrease due to the leakage current ILK flowing out from the output node NO through the first switching device **121**. Specifically, because the gate terminal and source terminal of the fifth NMOS transistor MN**5** of the leakage current compensation circuit **160** are short-circuited, the gate-source voltage of the fifth NMOS transistor MN**5** may always be maintained at 0 V. Accordingly, the turned-off state of the fifth NMOS transistor MN**5** may be maintained. That is, only the leakage compensation current ICP, which is the off leakage current, may flow from the drain terminal to the source terminal of the fifth NMOS transistor MN**5**. The leakage compensation current ICP may have an amount corresponding to the drain-source voltage of the fifth NMOS transistor MN**5**, that is, the difference VDD-VOSC between the supply voltage VDD and the oscillation voltage VOSC. In an example, by appropriately adjusting the ratio (channel width/channel length) of the fifth NMOS transistor MN**5** and the third NMOS transistor MN**3**, the leakage compensation current ICP may be the same as the leakage current ILK flowing through the first switching device **121**. The leakage compensation current ICP is supplied to the output node NO from the leakage current compensation circuit **160** through the output line to compensate for the decrease in the ramp current IRAMP by the leakage current ILK at the output node NO.

**[0068]** FIG. **13** is a waveform diagram illustrating the relationship between an oscillation signal output from an output node of a relaxation oscillation circuit included in an oscillator shown in FIG. **1**, leakage current, and a leakage compensation current. The following description may apply both when a charging operation is performed in the first RC circuit **111** and a discharging operation is performed in the second RC circuit **112** between a first time point T**1** and a second time point T**2** and when a discharging operation is performed in the first RC circuit **111** and a charging operation is performed in the second RC circuit **112** between a first time point T**1** and a second time point T**2**.

**[0069]** Referring to FIG. **13**, between the first time point T**1** and the second time point T**2**, when the first RC circuit **111** and the second RC circuit **112** perform a charging operation and a discharging operation, respectively, the oscillation voltage VOSC output from the output node NO of the relaxation oscillating circuit **100** may be the first oscillation voltage VOSC**1**, which is the voltage of the first node N**1** of the first RC circuit **111**. That is, from the first time point T**1** at which the first capacitor C**1** of the first RC circuit **111** begins to be charged to the second time point T**2** at which charging of the first capacitor C**1** is completed, the oscillation voltage VOSC from the output node NO may have a waveform that increases from 0 V to the first oscillation voltage VOSC**1**. In this process, as described with reference to FIG. **11**, the leakage current ILK corresponding to the oscillation voltage VOSC may flow from the output node NO to the ground voltage terminal through the second switching device **122**. Between the first time point T**1** and the second time point T**2**, the leakage current compensation circuit **160** may supply the leakage compensation current ICP corresponding to the difference VDD-VOSC between the supply voltage VDD and the oscillation voltage VOSC to the output node NO.

**[0070]** Between the first time point T**1** and the second time point T**2**, when the first RC circuit **111** and the second RC circuit **112** perform a discharging operation and a charging operation respectively, the oscillation voltage VOSC output from the output node NO of the relaxation oscillating circuit **100** may be the second oscillation voltage VOSC**2**, which is the voltage of the second node N**2** of the second RC circuit **112**. That is, from the first time point T**1** at which the second capacitor C**2** of the second RC circuit **112** begins to be charged to the second time point T**2** at which charging of the second capacitor C**2** is completed, the oscillation voltage VOSC from the output node NO may have a waveform that increases from 0 V to the second oscillation voltage VOSC**2**. In this process, as described with reference to FIG. **12**, the leakage current ILK corresponding to the oscillation voltage VOSC may flow from the output node NO to the ground voltage terminal through the first switching device **121**. Between the first time point T**1** and the second time point T**2**, the leakage current compensation circuit **160** may supply the leakage compensation current ICP corresponding to the difference VDD-VOSC between the supply voltage VDD and the oscillation voltage VOSC to the output node NO.

**[0071]** FIG. **14** is a circuit diagram illustrating an oscillator according to another embodiment of the present disclosure.

**[0072]** Referring to FIG. **14**, an oscillator **20** may include a relaxation oscillating circuit **300** and a voltage averaging feedback circuit **200**. The voltage averaging feedback circuit **200** is the same as described above with reference to FIG. **1**, so redundant description will be omitted below. The relaxation oscillating circuit **300** may include a first RC circuit **111**, a second RC circuit **112**, a first switching device **121**, a second switching device **122**, a first comparator **131**, a second comparator **132**, a latch circuit **140**, a first inverter **151**, a second inverter **152**, and a leakage current compensation circuit **360**. The first RC circuit **111**, the second RC circuit **112**, the first switching device **121**, the second switching device **122**, the first comparator **131**, the second comparator **132**, the latch circuit **140**, the first inverter **151**, and the second inverter **152** may be the same as described above with reference to FIG. **1**.

**[0073]** The leakage current compensation circuit **360** may be different from the leakage current compensation circuit (**160** in FIG. **1**) included in the oscillator (**10** in FIG. **1**) of FIG. **1**. Both circuits receive a supply voltage VDD but the leakage current compensation circuit **360** receives a control voltage VC output from the voltage averaging feedback circuit **200**. That is, a leakage compensation current ICP provided from the leakage current compensation circuit **360** to an output node NO may be generated by the difference between control voltage VC and oscillation voltage VOSC, which is the drain-source voltage of a fifth NMOS transistor MN**5**. A peak value of an oscillation voltage VOSC may vary depending on the process-voltage-temperature (hereinafter, referred to as "PVT") variation. The control voltage VC provided from the voltage averaging feedback circuit **200** has a value similar to the peak value of the oscillation voltage VOSC, so the leakage compensation current ICP may change in peak value according to the PVT variation that varies the oscillation voltage VOSC.

**[0074]** FIG. **15** is a waveform diagram illustrating a relationship between an oscillation signal output from an output node of a relaxation oscillating circuit included in an oscillator shown in FIG. **14**, a leakage current, and a leakage compensation current. The following description may apply both when a charging operation is performed in the first RC circuit **111** and a discharging operation is performed in the second RC circuit **112** between a first time point T**1** and a second time point T**2** and when a discharging operation is performed in the first RC circuit **111** and a charging operation is performed in the second RC circuit **112** between a first time point T**1** and a second time point T**2**.

**[0075]** Referring to FIG. **15**, between the first time point T**1** and the second time point T**2**, when the first RC circuit **111** and the second RC circuit **112** perform the charging operation and discharging operation respectively, the oscillation voltage VOSC output from the output node NO of the relaxation oscillating circuit **300** may be the first oscillation voltage VOSC**1**, which is the voltage of a first node N**1** of the first RC circuit **111**. That is, from the first time point T**1** at which a first capacitor C**1** of the first RC circuit **111** begins to be charged to the second time point T**2** at which the charging of the first capacitor C**1** is completed, the oscillation voltage VOSC from the output node NO may have a waveform that increases from 0 V to the first oscillation voltage VOSC**1**. In this process, as described with reference to FIG. **11**, the leakage current ILK corresponding to the oscillation voltage VOSC may flow from the output node NO to the ground voltage terminal through the second switching device **122**. Between the first time point T**1** and the second time point T**2**, the leakage current compensation circuit **360** may supply the leakage compensation current ICP corresponding to the difference VDD-VOSC of the supply voltage VDD and the oscillation voltage VOSC to the output node NO. As described with reference to FIG. **14**, the control voltage VC may have a magnitude similar to the peak value of the oscillation voltage VOSC.

**[0076]** Between the first time point T**1** and the second time point T**2**, when the first RC circuit **111** and the second RC circuit **112** perform the discharging operation and charging operation respectively, the oscillation voltage VOSC output from the output node NO of the relaxation oscillating circuit **300** may be the second oscillation voltage VOSC**2**, which is the voltage of the second node N2 of the second RC circuit **112**. That is, from the first time point T1 at which a second capacitor C2 of the second RC circuit **112** begins to be charged to the second time point T2 at which charging of the second capacitor C2 is completed, the oscillation voltage VOSC from the output node NO may have a waveform that increases from 0 V to the second oscillation voltage VOSC2. In this process, as described with reference to FIG. **12**, the leakage current ILK corresponding to the oscillation voltage VOSC may flow from the output node NO to the ground voltage terminal through the first switching device **121**. Between the first time point T1 and the second time point T2, the leakage current compensation circuit **360** may supply the leakage compensation current ICP corresponding to (VC-VOSC) to the output node NO.

**[0077]** FIG. **16** is a circuit diagram illustrating an oscillator according to yet another embodiment of the present disclosure.

**[0078]** Referring to FIG. **16**, an oscillator **30** may include a relaxation oscillating circuit **400** and a voltage averaging feedback circuit **200**. The voltage averaging feedback circuit **200** is the same as that described above with reference to FIG. **1**, so a redundant description will be omitted below. The relaxation oscillating circuit **400** may include a first capacitor circuit **411**, a second capacitor circuit **412**, a first switching device **121**, a second switching device **122**, a first comparator **131**, a second comparator **132**, a latch circuit **140**, a first inverter **151**, a second inverter **152**, and a leakage current compensation circuit **160**. The first switching device **121**, the second switching device **122**, the first comparator **131**, the second comparator **132**, the latch circuit **140**, the first inverter **151**, the second inverter **152**, and the leakage current compensation circuit **160** may be the same as components described above with reference to FIG. **1**. The relaxation oscillating circuit **400** in FIG. **16** may be different from the relaxation oscillating circuit (**100** in FIG. **1**) of the oscillator (**10** in FIG. **1**), in that the relaxation oscillating circuit (**100** in FIG. **1**) includes a first RC circuit (**111** in FIG. **1**) and a second RC circuit (**112** in FIG. **1**) while the relaxation oscillating circuit **400** of the oscillator **30** includes the first capacitor circuit **411** and the second capacitor circuit **412**.

**[0079]** The first capacitor circuit **411** may include a first current source CS1, a first capacitor C1, a first PMOS transistor MP1, and a first NMOS transistor MN1. The first current source CS1 may be coupled to a supply voltage VDD terminal to which the supply voltage VDD is applied through an input terminal, and may be coupled to a source terminal of the first PMOS transistor MP1 through an output terminal. A drain terminal of the first PMOS transistor MP1 may be coupled to a first node N1. A gate terminal of the first PMOS transistor MP1 may be coupled to a first output terminal Q of the latch circuit **140**. The first capacitor C1 may be coupled to the first node N1 and a ground voltage terminal. A drain terminal and a source terminal of the first NMOS transistor MN1 may be coupled to the first node N1 and the ground voltage terminal, respectively. A gate terminal of the first NMOS transistor MN1 may be coupled to the first output terminal Q of the latch circuit **140**.

**[0080]** When a first latch signal LAT1 of a logic "low" level is applied to the gate terminal of the first PMOS transistor MP1 and the gate terminal of the first NMOS transistor MN1, the first capacitor circuit **411** may perform a charging operation. A third PMOS transistor MP3 and a third NMOS transistor MN3 of the first switching device **121** may be turned on while the first capacitor circuit **411** performs the charging operation. As a current from the first current source CS1 flows into the first capacitor C1 through the first PMOS transistor MP1, the first capacitor C1 may be charged, and a voltage at the first node N1 may be pulled-up to a first oscillation voltage VOSC1. Because the first switching device **121** is in a turned-on state, the first oscillation voltage VOSC1 at the first node N1 may be applied to an output node NO. The first oscillation voltage VOSC1 may be output as an oscillation voltage VOSC through an output line of the relaxation oscillating circuit **400**. While the first capacitor circuit **411** performs the charging operation as described above with reference to FIG. **11**, the leakage current compensation circuit **160** may provide a leakage compensation current to the output node NO.

**[0081]** When the first latch signal LAT1 of a logic "high" level is applied to the gate terminal of the first PMOS transistor MP1 and the gate terminal of the first NMOS transistor MN1, the first capacitor circuit **411** may perform a discharging operation. The third PMOS transistor MP3 and the third NMOS transistor MN3 of the first switching device **121** may be turned off while the first capacitor circuit **411** performs the discharging operation. Because the first switching device **121** is in a turned-off state, the first capacitor C1 may discharge electric charges to the ground voltage terminal through the first NMOS transistor MN1. Accordingly, the voltage at the first node N1 may be pulled-down from the first oscillation voltage VOSC1 until the voltage reaches 0 V.

**[0082]** The second capacitor circuit **412** may include a second current source CS2, a second capacitor C2, a second PMOS transistor MP2, and a second NMOS transistor MN2. The second current source CS2 may be coupled to the supply voltage VDD terminal to which the supply voltage VDD is applied through an input terminal, and may be coupled to a source terminal of the second PMOS transistor MP2 through an output terminal. A drain terminal of the second PMOS transistor MP2 may be coupled to a second node N2. A gate terminal of the second PMOS transistor MP2 may be coupled to a second output terminal QB of the latch circuit **140**. The second capacitor C2 may be coupled to the second node N2 and the ground voltage terminal. A drain terminal and a source terminal of the second NMOS transistor MN2 may be coupled to the second node N2 and the ground voltage terminal, respectively. A gate terminal of the second NMOS transistor MN2 may be coupled to the second output terminal QB of the latch circuit.

**[0083]** When a second latch signal LAT2 of a logic "low" level is applied to the gate terminal of the second PMOS transistor MP2 and the gate terminal of the second NMOS transistor MN2, the second capacitor circuit **412** may perform a charging operation. While the second capacitor circuit **412** performs the charging operation, a fourth PMOS transistor MP4 and a fourth NMOS transistor MN4 of the second switching device **122** may be turned on. As current from the second current source CS2 flows into the second capacitor C2 through the second PMOS transistor MP2, the second capacitor C2 may be charged, and the voltage at the second node N2 may be pulled-up to a second oscillation voltage VOSC2. Because the second switching device **122** is in a turned-on state, the second oscillation voltage VOSC2 at the second node N2 may be applied to the output node NO. The second oscillation voltage VOSC2 may be output as the oscillation voltage VOSC through the output line of the relaxation oscillating circuit **400**. While the second capacitor circuit **412** performs the charging operation as described above with reference to FIG. **12**, the leakage current compensation circuit **160** may provide a leakage compensation current to the output node NO.

**[0084]** When the second latch signal LAT**2** of a logic "high" level is applied to the gate terminal of the second PMOS transistor MP**2** and the gate terminal of the second NMOS transistor MN**2**, the second capacitor circuit **412** may perform a discharging operation. While the second capacitor circuit **412** performs the discharging operation, the fourth PMOS transistor MP**4** and the fourth NMOS transistor MN**4** of the second switching device **122** may be turned off. Because the second switching device **122** is in a turned-off state, the second capacitor C**2** may discharge electric charges to the ground voltage terminal through the second NMOS transistor MN**2**. Accordingly, the voltage at the second node N**2** may be pulled-down from the second oscillation voltage VOSC**2** until the voltage reaches 0 V.

**[0085]** FIG. **17** is a circuit diagram illustrating an oscillator according to still another embodiment of the present disclosure.

**[0086]** Referring to FIG. **17**, an oscillator **40** may include a relaxation oscillating circuit **500** and a voltage averaging feedback circuit **200**. The voltage average feedback circuit **200** is the same circuit as described with reference to FIG. **1**, so a redundant description will be omitted below. The relaxation oscillating circuit **500** may include a first capacitor circuit **411**, a second capacitor circuit **412**, a first switching device **121**, a second switching device **122**, a first comparator **131**, a second comparator **132**, a latch circuit **140**, a first inverter **151**, a second inverter **152**, and a leakage current compensation circuit **360**. The first switching device **121**, the second switching device **122**, the first comparator **131**, the second comparator **132**, the latch circuit **140**, the first inverter **151**, and the second inverter **152** may be the same as described above and with reference to FIG. **1**. The first capacitor circuit **411** and the second capacitor circuit **412** may be the same as described above with reference to FIG. **16**. In addition, the leakage current compensation circuit **360** may be the same as described above with reference to FIG. **14**. That is, the relaxation oscillating circuit **500** of the oscillator **40** according to the present disclosure may be different from the relaxation oscillating circuit **100** of the oscillator **10** described above and with reference to FIG. **1**, in that the relaxation oscillating circuit **100** in FIG. **1** includes a first RC circuit **111** and a second RC circuit **112** while the relaxation oscillating circuit **500** of the oscillator **40** includes a first capacitor circuit **411** and a second capacitor circuit **412**. In addition, the leakage current compensation circuit **360** included in the relaxation oscillating circuit **500** of the oscillator **40** may be different from the leakage current compensation circuit **160** included in the oscillator **30** of FIG. **16**. The leakage current compensation circuit **160** receives a supply voltage VDD, while the leakage current compensation circuit **360** included in the relaxation oscillating circuit **500** receives a control voltage VC output from the voltage averaging feedback circuit **200**.

**[0087]** As described with reference to FIG. **14**, a leakage compensation current ICP provided from a leakage current compensation circuit **360** may be generated by a drain-source voltage (control voltage VC-oscillation voltage VOSC) of a fifth NMOS transistor MN**5**. The peak value of the oscillation voltage VOSC may vary depending on the process-voltage-temperature (PVT) variation. The control voltage VC provided from the voltage averaging feedback circuit **200** has a value similar to the peak value of the oscillation voltage VOSC, so the leakage compensation current ICP may correspond to a change in the peak value of the oscillation voltage VOSC, which may depend on PVT variation.

**[0088]** FIG. **18** is a circuit diagram illustrating an oscillator according to a further embodiment of the present disclosure.

**[0089]** Referring to FIG. **18**, an oscillator **50** may include a relaxation oscillating circuit **600** and a voltage averaging feedback circuit **200**. The voltage averaging feedback circuit **200** is the same as that described above with reference to FIG. **1**, so a redundant description will be omitted below. The relaxation oscillating circuit **600** may include a first RC circuit **611**, a second RC circuit **612**, a first switching device **621**, a second switching device **622**, a first comparator **131**, a second comparator **132**, a latch circuit **140**, a first inverter **151**, a second inverter **152**, and a leakage current compensation circuit **660**. The first comparator **131**, the second comparator **132**, the latch circuit **140**, the first inverter **151**, and the second inverter **152** may be the same as that described above and with reference to FIG. **1**.

**[0090]** The first RC circuit **611** may include a first NMOS transistor MN**1**, a first resistor R**1**, a second NMOS transistor MN**2**, a first capacitor C**1**, and a first PMOS transistor MP**1**. A source terminal of the first NMOS transistor MN**1** may be coupled to the ground voltage terminal. A drain terminal of the first NMOS transistor MN**1** may be coupled to the first resistor R**1**. A gate terminal of the first NMOS transistor MN**1** may be coupled to a first output terminal Q of the latch circuit **140**. One terminal of the first resistor R**1** may be coupled to the drain terminal of the first NMOS transistor MN**1**. The other terminal of the first resistor R**1** may be coupled to a first node N**1**. A source terminal of the second NMOS transistor MN**2** may be coupled to the ground voltage terminal. A drain terminal of the second NMOS transistor MN**2** may be coupled to the first node N**1**. A temperature compensation voltage VTC may be applied to a gate terminal of the second NMOS transistor MN**2**. One terminal of the first capacitor C**1** may be coupled to the first node N**1**. The other terminal of the first capacitor C**1** may be coupled to the supply voltage VDD terminal. A drain terminal of the first PMOS transistor MP**1** may be coupled to the first node N**1**. A source terminal of the first PMOS transistor MP**1** may be coupled to the supply voltage VDD terminal. A gate terminal of the first PMOS transistor MP**1** may be coupled to the first output terminal Q of the latch circuit **140**.

**[0091]** The second RC circuit **612** may include a third NMOS transistor MN**3**, a second resistor R**2**, a fourth NMOS transistor MN**4**, a second capacitor C**2**, and a second PMOS transistor MP**2**. A source terminal of the third NMOS transistor MN**3** may be coupled to the ground voltage terminal. A drain terminal of the third NMOS transistor MN**3** may be coupled to the second resistor R**2**. A gate terminal of the third NMOS transistor MN**3** may be coupled to a second output terminal QB of the latch circuit **140**. One terminal of the second resistor R**2** may be coupled to the drain terminal of the third NMOS transistor MN**3**. The other terminal of the second resistor R**2** may be coupled to a second node N**2**. A source terminal of the fourth NMOS transistor MN4 may be coupled to the ground voltage terminal. A drain terminal of the fourth NMOS transistor MN4 may be coupled to the second node N2. A temperature compensation voltage VTC may be applied to a gate terminal of the fourth NMOS transistor MN4. One terminal of the second capacitor C2 may be coupled to the second node N2. The other terminal of the second capacitor C2 may be coupled to the supply voltage VDD terminal. A drain terminal of the second PMOS transistor MP2 may be coupled to the second node N2. A source terminal of the second PMOS transistor MP2 may be coupled to the supply voltage VDD terminal. A gate terminal of the second PMOS transistor MP2 may be coupled to the second output terminal QB of the latch circuit **140**.

**[0092]** The first capacitor C1 of the first RC circuit **611** is coupled to the supply voltage VDD terminal, so the first capacitor C1 may be in a charged state. Accordingly, a first oscillation voltage VOSC1 may be applied to the first node N1. Similarly, the second capacitor C2 of the second RC circuit **612** is coupled to the supply voltage VDD terminal, so the second capacitor C2 may also be charged in an initial state, and accordingly, a second oscillation voltage VOSC2 may be applied to the second node N2. The first oscillation voltage VOSC1 at the first node N1 and the second oscillation voltage VOCS2 at the second node N2 may each have substantially the same magnitude as the supply voltage VDD.

**[0093]** The first switching device **621** may be a transmission gate including a third PMOS transistor MP3 and a fifth NMOS transistor MN5. A gate terminal of the third PMOS transistor MP3 may be coupled to an output terminal of the first inverter **151**. A source terminal and a drain terminal (or the drain terminal and the source terminal) of the third PMOS transistor MP3 may be coupled to the first node N1 and an output node NO, respectively. A gate terminal of the fifth NMOS transistor MN5 may be coupled to the first output terminal Q of the latch circuit **140**. A drain terminal and a source terminal (or the source terminal and the drain terminal) of the fifth NMOS transistor MN5 may be coupled to the first node N1 and the output node NO, respectively.

**[0094]** The second switching device **622** may be a transmission gate including a fourth PMOS transistor MP4 and a sixth NMOS transistor MN6. A gate terminal of the fourth PMOS transistor MP4 may be coupled to an output terminal of the second inverter **152**. A source terminal and a drain terminal (or the drain terminal and the source terminal) of the fourth PMOS transistor MP4 may be coupled to the second node N2 and the output node NO, respectively. A gate terminal of the sixth NMOS transistor MN6 may be coupled to the second output terminal QB of the latch circuit **140**. A drain terminal and a source terminal (or the source terminal and the drain terminal) of the sixth NMOS transistor MN6 may be coupled to the second node N2 and the output node NO, respectively.

**[0095]** The leakage current compensation circuit **660** may include a fifth PMOS transistor MP5 and a third resistor R3. A gate terminal and a source terminal of the fifth PMOS transistor MP5 may be coupled to an output line from the output node NO. Because the gate terminal and source terminal of the fifth PMOS transistor MP5 are short-circuited, the fifth PMOS transistor MP5 may maintain a turned-off state. A drain terminal of the fifth PMOS transistor MP5 may be coupled to one terminal of the third resistor R3. The other terminal of the third resistor R3 may be coupled to the output line of the voltage averaging feedback circuit **200**. Accordingly, the control voltage VC output from the voltage averaging feedback circuit **200** may be applied to the other terminal of the third resistor R3. In another embodiment, the leakage current compensation circuit **660** might not include the third resistor R3. In this case, the control voltage VC may be applied to the drain terminal of the fifth PMOS transistor MP5.

**[0096]** FIG. **19** is a circuit diagram illustrating an operation of an oscillator illustrated in FIG. **18**. In FIG. **19**, the same reference numerals as used in FIG. **18** indicate the same components.

**[0097]** Referring to FIG. **19**, when a first latch signal LAT1 of a logic "high" level and a second latch signal LAT2 of a logic "low" level are output through the first output terminal Q and the second output terminal QB of a latch circuit **140**, respectively, the first RC circuit **611** may perform a discharging process and the second RC circuit **612** may perform a charging process. In addition, the third PMOS transistor MP3 and the fifth NMOS transistor MN5 of the first switching device **621** may be turned on, and the fourth PMOS transistor MP4 and the sixth NMOS transistor MN6 of the second switching device **622** may be turned off. Accordingly, the oscillation voltage VOSC of the output node NO may become equal to the first oscillation voltage VOSC1 at the first node N1.

**[0098]** Although not illustrated in FIG. **19**, when the first latch signal LAT1 of a logic "low" level and the second latch signal LAT2 of a logic "high" level are output through the first output terminal Q and the second output terminal QB of the latch circuit **140**, respectively, the first RC circuit **611** may perform a charging process and the second RC circuit **612** may perform a discharging process. In addition, the third PMOS transistor MP3 and the fifth NMOS transistor MN5 of the first switching device **621** may be turned off, and the fourth PMOS transistor MP4 and the sixth NMOS transistor MN6 of the second switching device **622** may be turned on. Accordingly, the oscillation voltage VOSC at the output node NO may become equal to the second oscillation voltage VOSC2 at the second node N2.

**[0099]** When the first RC circuit **611** performs the discharging process and the second RC circuit **612** performs the charging process, that is, when the first latch signal LAT1 of a logic "high" level and the second latch signal LAT2 of a logic "low" level are output through the first output terminal Q and the second output terminal QB of the latch circuit **140**, respectively, the first NMOS transistor MN1 and the first PMOS transistor MP1 of the first RC circuit **611** may be turned on and turned off, respectively. On the other hand, the third NMOS transistor MN3 and the second PMOS transistor MP2 of the second RC circuit **612** may be turned off and turned on, respectively. The second NMOS transistor MN2 of the first RC circuit **611** and the fourth NMOS transistor MN4 of the second RC circuit **612** may both be turned on as the temperature compensation voltage VTC is applied to the gate terminals.

**[0100]** Due to the first oscillation voltage VOSC1 at the first node N1 of the first RC circuit **611**, a ramp current IRAMP may be divided into a first ramp current IR1 and a second ramp current IR2 through a first path and a second path, respectively, and flow to the ground voltage terminal. As a result, the voltage at the first node N1 of the first RC circuit **611** may be pulled-down from the first oscillation voltage VOSC1 until the voltage reaches 0 V. Here, the first path may be a path from the first node N1 through the first resistor R1 and the first NMOS transistor MN1, and the second path may be a path through the second NMOS transistor MN2. The temperature compensation voltage VTC applied to the gate terminal of the second NMOS transistor MN2 may vary in magnitude such that the amount of the second ramp current IR2 decreases in proportion to the increase in the first ramp current IR1, while the sum of the first ramp current IR1 and the second ramp current IR2 remains equal to the amount of the ramp current IRAMP. Accordingly, when the amount of the first ramp current IR1 is equal to the amount of the ramp current IRAMP, the second NMOS transistor MN2 of the first RC circuit **611** and the fourth NMOS transistor MN4 of the second RC circuit **612** may be turned off.

**[0101]** While the first RC circuit **611** performs a discharging process, the second RC circuit **612** may perform a charging process. Specifically, as the second latch signal LAT2 of a logic "low" level is output through the second output terminal QB of the latch circuit **140**, the second PMOS transistor MP2 of the second RC circuit **612** may be turned on, and the third NMOS transistor MN3 may be turned off. In addition, the fourth PMOS transistor MP4 and the sixth NMOS transistor MN6 of the second switching device **622** may be turned off. The fourth NMOS transistor MN4 of the second RC circuit **612** may be turned on as the temperature compensation voltage VTC is applied to the gate terminal. Under such conditions, the voltage at the second node N2 of the second RC circuit **612** may be pulled-up to the second oscillation voltage VOSC2.

**[0102]** In this way, when the first RC circuit **611** performs a discharging process and the second RC circuit **612** performs a charging process, a leakage current ILK may flow from the second node N2 to the output node NO through the second switching device **622**, which is in a turned-off state. Specifically, the gate-source voltage of the sixth NMOS transistor MN6 may be 0 V, and accordingly, an off leakage current corresponding to the drain-source voltage of the sixth NMOS transistor MN6 may be generated. The leakage current ILK may increase the amount of the ramp current IRAMP from the output node NO to the first node N1, and as a result, the time required for the discharging process in the first RC circuit **611** increases.

**[0103]** In order to suppress problems caused by the leakage current ILK, the leakage current compensation circuit **660** may generate a leakage compensation current ICP flowing out from the output node NO to the control voltage VC terminal. Specifically, because the gate terminal and source terminal of the fifth PMOS transistor MP5 included in the leakage current compensation circuit **660** are short-circuited, the fifth PMOS transistor MP5 may be maintained in a turned-off state. The gate-source voltage of the fifth PMOS transistor MP5 is 0 V, and accordingly, an off leakage current, that is, the leakage compensation current ICP may flow from the source terminal to the drain terminal of the fifth PMOS transistor MP5. The leakage compensation current ICP may be generated in response to the source-drain voltage of the fifth PMOS transistor MP5, that is, the difference between the oscillation voltage VOSC and the control voltage VC.

**[0104]** A limited number of possible embodiments for the present teachings have been presented above for illustrative purposes. Those of ordinary skill in the art will appreciate that various modifications, additions, and substitutions are possible. While this patent document contains many specifics, these should not be construed as limitations on the scope of the present teachings or of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments. Certain features that are described in this patent document in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

What is claimed is:

**1**. An oscillator comprising:

a relaxation oscillating circuit including a first resistor-capacitor (RC) circuit generating a first oscillation voltage at a first node and a second RC circuit generating a second oscillation voltage at a second node, and configured to output an oscillation voltage through an output line coupled to an output node; and

a voltage averaging feedback circuit configured to receive a reference voltage and the oscillation voltage to output a control voltage through an output terminal,

wherein the relaxation oscillating circuit includes a leakage current compensation circuit coupled to the output line and configured to provide a leakage compensation current to the output node.

**2**. The oscillator of claim **1**,

wherein the first RC circuit includes:

a first MOS transistor coupled to a supply voltage terminal to which a supply voltage is applied;

a first resistor coupled to the first MOS transistor and the first node;

a first capacitor between the first node and a ground voltage terminal to which a ground voltage is applied; and

a second MOS transistor coupled to the first capacitor in parallel between the first node and the ground voltage terminal, and

wherein the second RC circuit includes:

a third MOS transistor coupled to the supply voltage terminal;

a second resistor coupled to the third MOS transistor and the second node;

a second capacitor between the second node and the ground voltage terminal; and

a fourth MOS transistor coupled to the second capacitor in parallel between the second node and the ground voltage terminal.

**3**. The oscillator of claim **2**,

wherein the relaxation oscillating circuit further includes:

a first switching device coupled to the first node and the output node; and

a second switching device coupled to the second node and the output node, and

wherein the first switching device and the second switching device perform switching operations so that the first oscillation voltage and the second oscillation voltage are alternately output through the output line coupled to the output node.

**4**. The oscillator of claim **3**, wherein the relaxation oscillating circuit further includes:

a first comparator configured to compare the control voltage and the first oscillation voltage at the first node to output a first comparison signal;

a second comparator configured to compare the control voltage and the second oscillation voltage at the second node to output a second comparison signal; and

a latch circuit configured to receive the first comparison signal and the second comparison signal to output a first latch signal and a second latch signal for alternately turning on the first switching device and the second switching device.

**5**. The oscillator of claim **4**,

wherein the first switching device is a first transmission gate including a first P-channel type MOS (PMOS) transistor and a first N-channel type MOS (NMOS) transistor, and

wherein the second switching device is a second transmission gate including a second PMOS transistor and a second NMOS transistor.

**6**. The oscillator of claim **5**,

wherein a gate terminal of the first PMOS transistor receives the first latch signal,

wherein a gate terminal of the first NMOS transistor receives an inverted signal of the first latch signal,

wherein a gate terminal of the second PMOS transistor receives the second latch signal, and

wherein a gate terminal of the second NMOS transistor receives an inverted signal of the second latch signal.

**7**. The oscillator of claim **5**, wherein each of the first PMOS transistor, the first NMOS transistor, the second PMOS transistor, and the second NMOS transistor has low voltage threshold (LVT) characteristics or ultra-high voltage threshold (ULVT) characteristics.

**8**. The oscillator of claim **2**,

wherein the leakage current compensation circuit includes an N-channel type MOS (NMOS) transistor coupled to the supply voltage terminal and the output line, and

wherein a drain terminal of the NMOS transistor is coupled to the supply voltage terminal, and a gate terminal and a source terminal of the NMOS transistor are coupled to the output line.

**9**. The oscillator of claim **8**, wherein the leakage current compensation circuit further includes a resistor coupled to the supply voltage terminal and the drain terminal of the NMOS transistor.

**10**. The oscillator of claim **1**,

wherein the leakage current compensation circuit includes an N-channel type MOS (NMOS) transistor coupled to the output terminal of the voltage averaging feedback circuit and the output line, and

wherein a drain terminal of the NMOS transistor is coupled to the output terminal of the voltage averaging feedback circuit, and a gate terminal and a source terminal of the NMOS transistor are coupled to the output line.

**11**. The oscillator of claim **10**, wherein the leakage current compensation circuit further includes a resistor coupled to the output terminal of the voltage averaging feedback circuit and the drain terminal of the NMOS transistor.

**12**. An oscillator comprising:

a relaxation oscillating circuit including a first capacitor circuit generating a first oscillation voltage at a first node and a second capacitor circuit generating a second oscillation voltage at a second node, and configured to alternately output the first oscillation voltage and the second oscillation voltage through an output line coupled to an output node; and

a voltage averaging feedback circuit configured to receive a reference voltage and one of the first oscillation voltage and the second oscillation voltage and to output a control voltage through an output terminal,

wherein the relaxation oscillating circuit includes a leakage current compensation circuit coupled to the output line and configured to provide a leakage compensation current to the output node.

**13**. The oscillator of claim **12**,

wherein the first capacitor circuit includes:

a first current source coupled to a supply voltage terminal to which a supply voltage is applied;

a first MOS transistor coupled to the first current source and the first node;

a first capacitor between the first node and a ground voltage terminal to which a ground voltage is applied; and

a second MOS transistor coupled to the first capacitor in parallel between the first node and the ground voltage terminal, and

wherein the second capacitor circuit includes:

a second current source coupled to the supply voltage terminal;

a third MOS transistor coupled to the second current source and the second node;

a second capacitor between the third MOS transistor and the ground voltage terminal; and

a fourth MOS transistor coupled to the second capacitor in parallel between the second node and the ground voltage terminal.

**14**. The oscillator of claim **13**,

wherein the relaxation oscillating circuit further includes:

a first switching device coupled to the first node and the output node; and

a second switching device coupled to the second node and the output node, and

wherein the first switching device and the second switching device perform switching operations so that the first oscillation voltage and the second oscillation voltage are alternately output through the output line coupled to the output node.

**15**. The oscillator of claim **14**,

wherein the relaxation oscillating circuit further includes:

a first comparator configured to compare the control voltage and the first oscillation voltage at the first node to output a first comparison signal;

a second comparator configured to compare the control voltage and the second oscillation voltage at the second node to output a second comparison signal; and

a latch circuit configured to receive the first comparison signal and the second comparison signal to output a first latch signal and a second latch signal for alternately turning on the first switching device and the second switching device.

**16**. The oscillator of claim **15**,

wherein the first switching device is a first transmission gate including a first P-channel type MOS (PMOS) transistor and a first N-channel type MOS (NMOS) transistor, and wherein the second switching device is a second transmission gate including a second PMOS transistor and a second NMOS transistor.

**17**. The oscillator of claim **16**,

wherein a gate terminal of the first PMOS transistor receives the first latch signal,

wherein a gate terminal of the first NMOS transistor receives an inverted signal of the first latch signal,

wherein a gate terminal of the second PMOS transistor receives the second latch signal, and

wherein a gate terminal of the second NMOS transistor receives an inverted signal of the second latch signal.

**18**. The oscillator of claim **16**, wherein each of the first PMOS transistor, the first NMOS transistor, the second PMOS transistor, and the second NMOS transistor has low voltage threshold (LVT) characteristics or ultra-high voltage threshold (ULVT) characteristics.

**19**. The oscillator of claim **12**,

wherein the leakage current compensation circuit includes an N-channel type MOS (NMOS) transistor coupled to a supply voltage terminal and the output line, and

wherein a drain terminal of the NMOS transistor is coupled to the supply voltage terminal, and a gate terminal and a source terminal of the NMOS transistor are coupled to the output line.

**20**. The oscillator of claim **19**, wherein the leakage current compensation circuit further includes a resistor coupled to the supply voltage terminal and the drain terminal of the NMOS transistor.

**21**. The oscillator of claim **12**,

wherein the leakage current compensation circuit includes an N-channel type MOS (NMOS) transistor coupled to the output terminal of the voltage averaging feedback circuit and the output line, and

wherein a drain terminal of the NMOS transistor is coupled to the output terminal of the voltage averaging feedback circuit, and a gate terminal and a source terminal of the NMOS transistor are coupled to the output line.

**22**. The oscillator of claim **21**, wherein the leakage current compensation circuit further includes a resistor coupled to the output terminal of the voltage averaging feedback circuit and the drain terminal of the NMOS transistor.

**23**. An oscillator comprising:

a relaxation oscillating circuit including a first resistor-capacitor (RC) circuit generating a first oscillation voltage at a first node and a second RC circuit generating a second oscillation voltage at a second node, and configured to alternately output the first oscillation voltage and the second oscillation voltage through an output line coupled to an output node; and

a voltage averaging feedback circuit configured to receive a reference voltage and one of the first oscillation voltage and the second oscillation voltage and to output a control voltage through an output terminal,

wherein the relaxation oscillating circuit includes a leakage current compensation circuit coupled to the output line and configured to counteract a leakage current from the output node.

**24**. The oscillator of claim **23**,

wherein the first RC circuit includes:

a first MOS transistor coupled to a ground voltage terminal to which a ground voltage is applied;

a first resistor coupled to the first MOS transistor and the first node;

a first capacitor between the first node and a supply voltage terminal to which a supply voltage is applied; and

a second MOS transistor coupled to the first capacitor in parallel between the first node and the supply voltage terminal, and

wherein the second RC circuit includes:

a third MOS transistor coupled to the ground voltage terminal;

a second resistor coupled to the third MOS transistor and the second node;

a second capacitor between the second node and the supply voltage terminal; and

a fourth MOS transistor coupled to the second capacitor in parallel between the second node and the supply voltage terminal.

**25**. The oscillator of claim **24**,

wherein the first RC circuit further includes a fifth MOS transistor coupled to the first MOS transistor in parallel between the ground voltage terminal and the first node, and

wherein the second RC circuit further includes a sixth MOS transistor coupled to the third MOS transistor in parallel between the ground voltage terminal and the second node.

**26**. The oscillator of claim **25**, wherein a temperature compensation voltage generated through a temperature compensation process is applied to a gate terminal of the fifth MOS transistor and a gate terminal of the sixth MOS transistor.

**27**. The oscillator of claim **25**,

wherein the relaxation oscillating circuit further includes:

a first switching device coupled to the first node and the output node; and

a second switching device coupled to the second node and the output node, and

wherein the first switching device and the second switching device perform switching operations so that the first oscillation voltage and the second oscillation voltage are alternately output through the output line coupled to the output node.

**28**. The oscillator of claim **27**, wherein the relaxation oscillating circuit further includes:

a first comparator configured to compare the control voltage and the first oscillation voltage at the first node to output a first comparison signal;

a second comparator configured to compare the control voltage and the second oscillation voltage at the second node to output a second comparison signal; and

a latch circuit configured to receive the first comparison signal and the second comparison signal to output a first latch signal and a second latch signal for alternately turning on the first switching device and the second switching device.

**29**. The oscillator of claim **28**,

wherein the first switching device is a first transmission gate including a first P-channel type MOS (PMOS) transistor and a first N-channel type MOS (NMOS) transistor, and

wherein the second switching device is a second transmission gate including a second PMOS transistor and a second NMOS transistor.

**30**. The oscillator of claim **29**,

wherein a gate terminal of the first PMOS transistor receives the first latch signal,

wherein a gate terminal of the first NMOS transistor receives an inverted signal of the first latch signal,

wherein a gate terminal of the second PMOS transistor receives the second latch signal, and

wherein a gate terminal of the second NMOS transistor receives an inverted signal of the second latch signal.

**31**. The oscillator of claim **29**, wherein each of the first PMOS transistor, the first NMOS transistor, the second PMOS transistor, and the second NMOS transistor has low voltage threshold (LVT) characteristics or ultra-high voltage threshold (ULVT) characteristics.

**32**. The oscillator of claim **23**,

wherein the leakage current compensation circuit includes a P-channel type MOS (PMOS) transistor coupled to the output terminal of the voltage averaging feedback circuit and the output line, and

wherein a drain terminal of the PMOS transistor is coupled to the output terminal of the voltage averaging feedback circuit, and a gate terminal and a source terminal of the PMOS transistor are coupled to the output line.

**33**. The oscillator of claim **32**, wherein the leakage current compensation circuit further includes a resistor coupled to the output terminal of the voltage averaging feedback circuit and the drain terminal of the PMOS transistor.

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